Downlink Beamforming for Cell-Free ISAC: A Fast Complex Oblique Manifold Approach

Shayan Zargari, Diluka Galappaththige, *Member, IEEE*, Chintha Tellambura, *Fellow, IEEE*, and Geoffrey Ye Li, *Fellow, IEEE,*

***Abstract*—Cell-free integrated sensing and communication (CF-ISAC) systems are just emerging as an interesting technique for future communications. Such a system comprises several multiple-antenna access points (APs), serving multiple single-antenna communication users and sensing targets. However, efficient beamforming designs that achieve high precision and robust performance in densely populated networks are lacking. This paper proposes a new beamforming algorithm by exploiting the inherent Riemannian manifold structure. The aim is to maximize the communication sum rate while satisfying sensing beampattern gains and per AP transmit power constraints. To address this constrained optimization problem, a highly efficient augmented Lagrangian model-based iterative manifold optimization for the CF-ISAC (ALMCI) algorithm is developed. This algorithm exploits the geometry of the proposed problem and uses a complex oblique manifold. Conventional convex-concave procedure (CCPA) and multidimensional complex quadratic transform (MCQT)-SCA algorithms are also developed as comparative benchmarks. The ALMCI algorithm significantly outperforms both of these. For example, with 16 APs having 12 antennas and $30\,\mathrm{dBm}$ transmit power each, our proposed ALMCI algorithm yields $22.7\,\%$ and $6.7\,\%$ sum rate gains over the CCPA and MCQT-SCA algorithms, respectively. In addition to improvement in communication capacity, the ALMCI algorithm achieves superior beamforming gains and reduced complexity.**

***Index Terms*—Integrated sensing and communication, transmit beamforming, manifold algorithm.**

I. Introduction

CELL-FREE (CF) integrated sensing and communication (ISAC) systems are emerging and receiving attention [1]–[7] recently. Their applications include the Internet of Things (IoT), autonomous vehicles, indoor localization, and tracking [1]. Transmit beamforming at the access points (APs) is critical to achieve optimal sensing and communication performance.

The CF architecture with APs for ISAC (CF-ISAC) offers significant advantages over single-AP or base station (BS)-based ISAC systems [2]–[7]. Distributed APs enable cooperative transmission and reception strategies, minimizing inter-cell and co-channel interference to ensure seamless service and uniform quality-of-service regardless of user location [8]–[10]. CF-ISAC also manages communication-sensing interference through centralized beamforming optimization [2]–[7].

From a sensing perspective, distributed APs provide multiple observation angles for multi-static sensing, improving accuracy, blockage robustness, and parameter estimation. This enables enhanced target detection and localization, even in challenging scenarios. Unlike traditional mono-static ISAC systems that rely on complex FD transceivers with self-interference cancellation [11], [12], CF-ISAC eliminates such hardware requirements, making it a highly effective paradigm for achieving superior performance in communication and sensing. CF-ISAC performance hinges on effective AP beamforming, yet optimizing it presents significant challenges. APs must cooperatively meet user rate demands, minimize multi-user interference, and maximize beam gains for target detection. Traditional methods like conjugate beamforming often fall short, degrading sensing performance. Advanced CF-ISAC beamforming strategies are essential to address these issues [2]–[7].

A. Previous Contributions

CF-ISAC beamforming designs [2]–[7] are at an embryonic stage. Existing studies exploit several classical techniques, such as the successive convex approximation (SCA), max-min fairness, and semi-definite relaxation (SDR). Specifically, [2] develops an SCA algorithm based on minimizing the sensing beampattern gain matching mean-squared error. The AP beamforming in [3] is designed based on a max-min fairness formulation to maximize the sensing signal-to-noise ratio (SNR) of a single target while adhering to a communication signal-to-interference-plus-noise ratio (SINR) constraint of multiple users. References [4] uses SDR to optimize beamforming while satisfying communication SINR restrictions. The beamforming and receive filtering in [5] maximize the sum of sensing SINR. In [6], APs transmit beamforming is designed using a convex-concave procedure algorithm (CCPA) based on SDR and SCA to maximize the sensing SINR. Reference [7] proposes hybrid beamforming at the APs using a block coordinate descent algorithm to maximize the weighted sum rate while maintaining beampattern similarity for sensing.

These prior algorithms can be classified into two types, namely (i) CCPA based on SDR and SCA (Section IV-C), and (ii) fractional programming (FP)-based multidimensional complex quadratic transform (MCQT) algorithms (Section IV-D).

These approximations, linearization, and relaxation can lead to performance losses. Moreover, the algorithms may struggle to scale in densely populated networks with high computational demands. As user numbers or AP antennas increase, they require more resources, longer execution times, and additional iterations to converge, posing challenges in dynamic, real-time environments.

B. Manifold Optimization

Manifold optimization (MO) differs from standard optimization techniques by exploiting the problem's inherent geometric

S. Zargari, D. Galappaththige, and C. Tellambura with the Department of Electrical and Computer Engineering, University of Alberta, Edmonton, AB, T6G 1H9, Canada (e-mail: {zargari, diluka.lg, ct4}@ualberta.ca).

G. Y. Li is with the ITP Lab, the Department of Electrical and Electronic Engineering, Imperial College London, SW7 2BX London, U.K.(e-mail: geoffrey.li@imperial.ac.uk).

structure. Instead of searching through the entire space, MO restricts the search to a manifold. Thus, MO navigates the manifold's tangent spaces that locally resemble $\mathbb{R}^n$. These manifolds, such as lines, circles, planes, spheres, and matrix groups, have unique geometric properties. More details on MO can be found in [13]–[16] and references therein.

However, the existing CF-ISAC algorithms do not leverage the problem's geometric structure, which could enhance performance and convergence in high-dimensional scenarios if used properly. In contrast, the MO method is less sensitive to initialization and more scalable and therefore offers promising alternatives for large-scale, dynamic wireless networks [13], [15]. MO algorithms differ from conventional methods in gradient calculation, constraint handling, optimization path, and complexity [13]–[16]. While conventional techniques may traverse infeasible regions needing correction, MO searches inherently stay on the manifold, ensuring a more direct and efficient trajectory [13].

Thus, MO algorithms have recently been developed for several communication and radar problems [15], [17]–[20]. For instance, a data detection algorithm for massive multi-user multiple-input and multiple-output (MIMO) systems has been developed in [17]. In [18], a Riemannian conjugate gradient (RCG)-based multi-cast beamforming algorithm is presented over a complex oblique manifold to maximize the minimum SINR across users in a coordinated multi-cell network. The transmit beampattern design for MIMO radar systems in [20] converts the initial non-convex problem into an unconstrained quadratic Lagrange multiplier problem on a complex circle manifold. Reference [19] solves the constant envelope precoding problem in multi-user massive MIMO systems on a complex circle manifold. Our paper [15] presents an innovative augmented Lagrangian manifold optimization (ALMO) transmit beamforming algorithm to maximize the communication sum rate while ensuring the ISAC sensing beam gain requirements. The previous MO methods are confined to specific system setups. The developed approach can readily accommodate multiple constraints, making it a more universal solution.

C. Our Contribution

The main contribution is developing an MO framework to improve CF-ISAC performance. Consider a generalized CF-ISAC system with multi-antenna APs, multiple communication users, and multiple sensing targets (Fig. 1). The goal is to design an optimal transmit beamforming strategy for the AP that maximizes the sum rate. This optimization is subject to constraints on the sensing beampattern gain and the per-AP transmit power limits. Since this problem is non-convex, a novel optimization algorithm is required.

The main contributions are summarized as follows:

- An innovative CF-ISAC beamforming algorithm is proposed based on a search restricted to a complex oblique manifold and by augmenting the objective function with a Lagrangian penalty factor [13], resulting in highly efficient augmented Lagrangian model (ALM)-based iterative MO for the CF-ISAC (ALMCI) algorithm.
- Following our prior work on traditional ISAC [15], the proposed framework combines the dual objectives of maximizing the sum communication rate while meeting the sensing beampattern gain requirements. Nonetheless, this is the first implementation of this method to CF-ISAC, innovatively addressing the non-convex optimization challenges inherent in CF-ISAC beamforming.
- The proposed ALMCI surpasses the state-of-the-art FP/SDR/SCA optimization techniques regarding communication sum rate and sensing beampattern gain. For instance, it respectively achieves gains of $22.7\,\%$ and $6.7\,\%$ in terms of sum rate for 16 APs with 12 antennas and $30\,\mathrm{dBm}$ transmit power each compared to CCPA and MCQT-SCA methods. Moreover, it also provides improved sensing beampattern gains while attenuating sidelobes.
- Remarkably, such performance gains are achieved while reducing the computational complexity and algorithm execution time. For example, ALMCI is 13 and 27 times faster than CCPA and MCQT-SCA, respectively, for four APs with 16 antennas and at $30\,\mathrm{dBm}$ transmit power each. Moreover, the speed-up factor increases with the number of AP antennas. For instance, with 28 AP antennas, ALMCI is faster by 60 and 34 times than the other two. Thus, the speed-up factor increases exponentially with the size of the network.
- Reduced computational time and memory usage can reduce energy consumption. For example, with four APs with 16 antennas, ALMCI uses $99.75\,\%$ reduced memory compared to CCPA. Similarly, compared to MCQT-SCA, ALMCI yields a $25.21\,\%$ memory saving. Efficient computations, particularly in large-scale networks, result in less energy consumption. Thus, ALMCI may help to reduce the excessive resource consumption for large-scale CF-ISAC networks.

Notation: $\mathbf{A}^{\mathrm{H}}$ and $\mathbf{A}^{\mathrm{T}}$ are the Hermitian (conjugate transpose) and the transpose of matrix $\mathbf{A}$. $\mathbf{I}_M$ denotes the M-by-M identity matrix. The Euclidean norm, absolute value, real part, and expectation operators are denoted by $\|\cdot\|$ and $|\cdot|$, $\Re(\cdot)$, and $\mathbb{E}\{\cdot\}$, respectively. Also, $\otimes$ denotes the Kronecker product. A circularly symmetric complex Gaussian (CSCG) random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}$ is denoted by $\sim \mathcal{CN}(\boldsymbol{\mu}, \mathbf{C})$. Besides, $\mathbb{C}^{M\times N}$ and $\mathbb{R}^{M\times 1}$ represent $M \times N$ dimensional complex matrices and $M \times 1$ dimensional real vectors, respectively. The $\mathrm{unit}(\mathbf{a})$ function normalizes each element of vector $\mathbf{a}$. The Hadamard product $\mathbf{A} \circ \mathbf{B}$ denotes element-wise multiplication, and $\mathrm{clip}_{[a,b]}(x)$ limits x to the interval $[a, b]$. $\mathrm{diag}(\cdot)$ is the diagonal matrix operator. The operation $\mathrm{unt}(\mathbf{a}) = \left[\frac{a_1}{|a_1|}, \ldots, \frac{a_n}{|a_n|}\right]$. Sets $\mathcal{K} \triangleq \{1, \ldots, K\}$, $\mathcal{N} \triangleq \{1, \ldots, N\}$, $\mathcal{M}_a \triangleq \{1, \ldots, M\}$, and $\mathcal{K}_k \triangleq \mathcal{K} \setminus k$ define indices, while $\mathcal{O}$ denotes big-O notation.

II. System, Channel, and Signal Models

This section presents the system, channel, and signal models, including the communication architecture, channel assumptions, mathematical representations, and the structure of

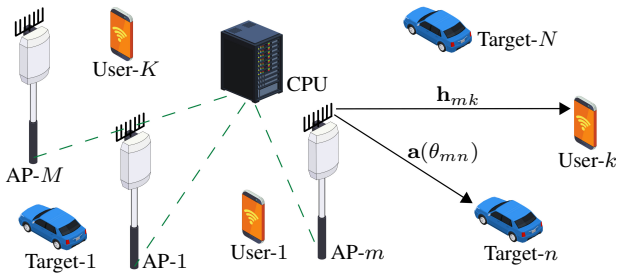


Fig. 1: A CF-ISAC system model.

transmitted and received signals with key parameters and variables.

A. System and Channel Models

In Fig. 1, the CF-ISAC system comprises M APs, each equipped with L antennas, K single-antenna communication users, and N sensing targets. The AP has a uniform linear array (ULA) of antennas spaced at half-wavelength intervals, facilitating precise communication and sensing [21]. The APs are connected to a central processing unit (CPU) via fronthaul/back-haul links, allowing APs to jointly and coherently communicate with users while sensing targets. This can mitigate inter-user interference, providing a uniformly quality service to all communication users [8]–[10], [22].

Each AP has digital beamforming, with each antenna element connected to an independent RF chain, allowing simultaneous transmission of communication and sensing waveforms to serve both users and targets efficiently. A CPU coordinates joint design and processing at the APs, ensuring full synchronization across all APs [8].

This work considers a conventional CF system with infinite-capacity fronthaul/backhaul links and perfect synchronization across all nodes [8]–[10], [22]. These assumptions, commonly used to facilitate theoretical analysis and algorithm design [3]–[5], simplify the study. However, deviations from these assumptions significantly impact performance. For instance, reducing fronthaul capacity per AP from $80\,\mathrm{bps/Hz}$ to $40\,\mathrm{bps/Hz}$ can lower the system sum rate by approximately $30\,\%$ [23]. Similarly, synchronization errors (e.g., time and phase misalignment) between APs degrade SINR at users, causing interference and loss of coherence in the received signal [24]. These limitations are expected to also hinder CF-ISAC systems with added sensing functionality. Investigating their effects is a promising research direction for practical CF-ISAC implementation. However, this work adopts standard assumptions to focus on the primary objective: developing and evaluating a novel, low-complexity beamforming algorithm for a generalized CF-ISAC system.

The communication environment is modeled using block flat-fading channels [15]. Within each fading block, channel vector $\mathbf{h}_{mk} \in \mathbb{C}^{L\times 1}$ represents the link from the m-th AP to the k-th user, and the vector $\mathbf{a}(\theta_{mn}) \in \mathbb{C}^{L\times 1}$ represents the channel from the m-th AP to the n-th target. Specifically, the communication channels are defined as

$$\mathbf{h}_{mk} = \zeta_{\mathbf{h}_{mk}}^{1/2}\tilde{\mathbf{h}}_{mk},\ \forall k, m, \tag{1}$$

where $\zeta_{\mathbf{h}_{mk}}$ denotes the large-scale path-loss and remains constant over several coherence intervals, and $\tilde{\mathbf{h}}_{mk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_L)$ represents the small-scale Rayleigh fading and follows a complex normal distribution.

To model the sensing channels between the APs and the targets, the echo signal representation in MIMO radar systems is adopted [21]. The line-of-sight (LoS) channels are thus represented using the transmit array steering vectors directed towards θ_{mn}, where θ_{mn} is the n-th target's direction from the m-th AP for the x-axis of the coordinate system. Specifically, channel vector $\mathbf{a}(\theta_{mn})$ is expressed as [21]:

$$\mathbf{a}(\theta_{mn}) = \sqrt{\frac{\zeta_{mn}}{L}}\left[1, e^{j\pi\sin(\theta_{mn})}, \ldots, e^{j\pi(L-1)\sin(\theta_{mn})}\right]^{\mathrm{T}},\ \forall n, m, \tag{2}$$

where ζ_{mn} is the is the path loss between the m-th AP and the n-th target.

Remark 1. *The following assumptions are standard and widely used:*

(i) θ_{mn} is assumed to be pre-estimated known by the CPU for beamforming design, which can be obtained using past scanning [21], [25], [26],

(ii) A separate channel estimation phase is employed before communication and sensing to ensure channel state information (CSI) is available for beamforming design.

(iii) The APs and the users are connected via a controlled link, exchanging necessary commands [27].

CF systems use the time division duplex (TDD) mode channel estimation. With this, the CSI can be readily estimated with algorithms, such as least squares and minimum mean-squared error (MMSE) estimators [28]–[30]. Thus, the AP and users are assumed to have CSI. This assumption is widely used [3]–[5]. Conversely, the control links operate separately from the conventional communication link and facilitate the exchange of essential commands and system information required for coordination between users and APs [27]. In particular, the control links enable the exchange of control information such as CSI, beamforming weights, and synchronization commands [27]. These messages are typically short, lightweight, and low-rate, minimizing overhead while ensuring seamless coordination and efficient system operation [27].

B. Transmission Model

The m-th AP transmit signal $\mathbf{x}_m \in \mathbb{C}^{L\times 1}$ serves both communication and sensing and is given by [15], [31]

$$\mathbf{x}_m = \sum_{k\in\mathcal{K}} \mathbf{v}_{mk} q_k,\ \forall m, \tag{3}$$

where $q_k \in \mathbb{C}$ is the data symbol for the k-th user with $\mathbb{E}\{|q_k|^2\} = 1$. The m-th AP beamforming vector for the k-th user, $\mathbf{v}_{mk} \in \mathbb{C}^{L\times 1}$, helps to communicate with the k-th user while achieving improved sensing beampattern gain on target directions to maximize the radar's signal strength [31]–[33]. The received signal at the k-th user can be expressed as

$$y_k = \sum_{m\in\mathcal{M}_a} \mathbf{h}_{mk}^{\mathrm{H}}\mathbf{v}_{mk}q_k + \sum_{m\in\mathcal{M}_a}\sum_{i\in\mathcal{K}_k} \mathbf{h}_{mk}^{\mathrm{H}}\mathbf{v}_{mi}q_i + n_k,\ \forall k, \tag{4}$$

where $n_k \sim \mathcal{CN}(0, \sigma^2)$ denotes the k-th user additive white Gaussian noise (AWGN). To achieve the signal model in (4), clutter rejection techniques can mitigate reflected interference from the targets and surrounding environment [34].

Remark 2. *While the APs can use separate beams for sensing, they also cause severe interference for communication users. To mitigate this, a dual-purpose beamforming strategy is employed to simultaneously serve communication and sensing purposes. It reduces the interference on communication users and maximizes the system's efficiency by using a single beam to handle both functionalities [31]–[33].*

Before proceeding to the problem formulation, several useful notations are introduced. First, all the beamforming vectors of the m-th AP for $m \in \mathcal{M}_a$ are organized into a single vector $\mathbf{v}_m = [\mathbf{v}_{m1}^{\mathrm{T}}, \ldots, \mathbf{v}_{mK}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^{LK\times 1}$. Second, the matrix $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_M] \in \mathbb{C}^{LK\times M}$, encapsulating beamforming vectors of all the APs for all the users, is constructed. Third, to effectively manage user and AP selections, selection matrices are introduced: $\mathbf{E}_k = [\mathbf{0}, \ldots, \mathbf{I}_L, \ldots, \mathbf{0}] \in \mathbb{R}^{L\times LK}$ for users, where the k-th block containing $L \times L$ identity matrix, i.e., $\mathbf{I}_L$, and $\mathbf{D} = \mathbf{I}_M \in \mathbb{R}^{M\times M}$ for APs.

III. Joint Communication and Sensing

This section focuses on deriving user communication rates and transmit beampattern gains for the targets.

A. Communication Performance

Here, the rates at which users decode their data using the AP signal are specified. From (4) and using the introduced notations, the SINR for the k-th user is expressed as

$$\gamma_k = \frac{|\sum_{m\in\mathcal{M}_a} \mathbf{h}_{mk}^{\mathrm{H}} \mathbf{E}_k \mathbf{V} \mathbf{D}_m|^2}{\sum_{i\in\mathcal{K}_k} |\sum_{m\in\mathcal{M}_a} \mathbf{h}_{mk}^{\mathrm{H}} \mathbf{E}_i \mathbf{V} \mathbf{D}_m|^2 + \sigma^2}, \ \forall k, \tag{5}$$

where $\mathbf{D}_m$ represents the m-th column of matrix $\mathbf{D}$. From (5), the communication rate of the k-th user can be approximately represented as

$$\mathcal{R}_k = \log_2(1 + \gamma_k), \ \forall k. \tag{6}$$

The rate quantifies the effectiveness of data decoding by users.

B. Sensing Performance

The transmit beampattern gain is a widely used metric for evaluating sensing signal design [35]. Beampattern gains, which reflect the power distribution of the transmit signal across different sensing angles θ, can enhance detection, recognition, and sensing accuracy [35].

The joint transmit beampattern gains can be used to achieve superior sensing outcomes, such as improved detection, range/Doppler/angle estimation, and tracking [36]. This objective implies the joint processing of sensing signals across the CPU's M sensing receivers. The sensing beampattern gain for the n-th target is given by [36]

$$p(\theta_n) = \mathbb{E}\left\{ \sum_{m\in\mathcal{M}_a} |\mathbf{a}^{\mathrm{H}}(\theta_{mn}) \mathbf{x}_m|^2 \right\} \tag{7}$$

$$= \sum_{m\in\mathcal{M}_a} \sum_{k\in\mathcal{K}} \mathbf{a}^{\mathrm{H}}(\theta_{mn}) \mathbf{E}_k \mathbf{V} \mathbf{D}_m (\mathbf{E}_k \mathbf{V} \mathbf{D}_m)^{\mathrm{H}} \mathbf{a}(\theta_{mn}), \ \forall n.$$

As a result, this measure is designed specifically to meet the target sensing requirements. A uniformly distributed beampattern is most efficient if the target directions are unknown. Conversely, in applications where the potential directions of targets are approximately known, the beampattern gain should be maximized towards these directions, enhancing the efficiency of target detection and tracking [35].

Other standard measures for evaluating sensing performance are transmitted beampattern mean-squared error (MSE) [37] and sensing rate [21]. Specifically, the beampattern MSE minimizes the discrepancy between the sensing and ideal beampatterns. However, it requires the knowledge of the ideal beampattern, which may not be available. Conversely, sensing rate accounts for both transmit and receiver beampatterns, indicating how much environmental information can be extracted from a specific target's reflected signal. Yet, this demands the transmitter to operate in FD mode, significantly impacting performance owing to SI. Thus, we employ beampattern gain as the viable performance metric [35], [36]. Nevertheless, our approach can easily handle beampattern MSE and sensing rate.

Our ISAC design prioritizes two main metrics: (1) the communication SINR and (2) the sensing beampattern gain. The former ensures high symbol detection accuracy and minimizes communication errors. On the other hand, sensing efficiency mainly depends on the beampattern gain, significantly affecting the target detection probability. A well-designed sensing beampattern gain helps achieve effective target recognition and overall sensing success [35].

C. Problem Formulation

As mentioned above, the objective is to maximize the communication sum rate for users while meeting sensing targets and per-AP transmit power constraints. The problem can thus be stated as follows:

$$\text{(P1)}: \ \max_{\mathbf{V}} \ \sum_{k\in\mathcal{K}} \log_2(1 + \gamma_k), \tag{8a}$$

$$\text{s.t.} \ \ p(\theta_n) \geq \Gamma_n^{\mathrm{th}}, \ \forall n, \tag{8b}$$

$$\sum_{k\in\mathcal{K}} \|\mathbf{E}_k \mathbf{V} \mathbf{D}_m\|^2 \leq p_{\max}, \ \forall m. \tag{8c}$$

where (8b) guarantees the sensing beampattern gain over the threshold of the n-th target, Γ_n^{th}, and (8c) indicates the maximum allowable transmit power of the m-th AP, as $p_{\max}$. This optimization problem aligns with the envisioned capabilities of 6G ISAC systems [38], [39].

IV. Optimal Beamforming

This section develops the MO solution to (P1), utilizing FP and MO to determine optimal $\mathbf{v}_{mk} \in \mathbb{C}^{L\times 1}$ [13], [16]. The key challenge of (P1) is its non-convex objective function.

Due to the objective's non-convex sum-log terms, there is no known convex reformulation for the problem (P1). However, the Lagrangian dual transform can convert the objective to a sum-of-ratios form [40, *Theorem 3*]. By introducing auxiliary

variables μ_k to replace SINR term γ_k in (8a) such that $\mu_k \leq \gamma_k$, (P1) is reformulated as [40]

$$\text{(P2)}: \max_{\mathbf{V},\boldsymbol{\mu}} \; f(\mathbf{V},\boldsymbol{\mu}) = \frac{1}{\ln(2)} \sum\nolimits_{k\in\mathcal{K}} \ln(1+\mu_k) + \frac{1}{\ln(2)} \sum\nolimits_{k\in\mathcal{K}} \left(-\mu_k + \frac{(1+\mu_k)\gamma_k}{1+\gamma_k}\right), \quad \text{(9a)}$$

$$\text{s.t.} \quad \text{(8b)} - \text{(8c)}, \quad \text{(9b)}$$

where $\boldsymbol{\mu} = [\mu_1, \ldots, \mu_K]$ represents the vector of auxiliary variables introduced by FP. Revised (P2) embodies a dual-phase optimization challenge, comprising (i) an outer optimization of $\mathbf{V}$ with a static $\boldsymbol{\mu}$ and (ii) an inner optimization of $\boldsymbol{\mu}$ with a static $\mathbf{V}$. To address (P2), an iterative strategy is deployed, alternately refining $\mathbf{V}$ and $\boldsymbol{\mu}$ until the objective function stabilizes.

Note that the original problem is recast into an equivalent form, in which $\mathbf{V}$ resolves (P1) if and only if it also solves (P2) [40, *Theorem 3*]. Moreover, it is established that the optimal objective values of (P1) and (P2) are identical [40].

A. Optimization of $\boldsymbol{\mu}$ with Fixed $\mathbf{V}$

The iterative optimization process first updates auxiliary variable $\boldsymbol{\mu}$ based on the previously determined values of $\mathbf{V}$. Notably, function $f(\mathbf{V},\boldsymbol{\mu})$ becomes concave and differentiable over $\boldsymbol{\mu}$ when $\mathbf{V}$ remains constant, allowing for an optimal determination of $\boldsymbol{\mu}$ by setting $\frac{\partial f(\mathbf{V},\boldsymbol{\mu})}{\partial \mu_k} = 0$ for each component. The resulting update rule is thus given as [41]

$$\mu_k^* = \gamma_k, \; \forall k. \quad \text{(10)}$$

For given $\boldsymbol{\mu}$, one can eliminate the constant terms with respect to $\mathbf{V}$ in (P2) to further simplify the optimization as

$$\text{(P3)}: \max_{\mathbf{V}} \sum_{k\in\mathcal{K}} \frac{\tilde{\mu}_k |\sum_{m\in\mathcal{M}_a} \mathbf{h}_{mk}^{\mathrm{H}} \mathbf{E}_k \mathbf{V} \mathbf{D}_m|^2}{\sum_{i\in\mathcal{K}} |\sum_{m\in\mathcal{M}_a} \mathbf{h}_{mk}^{\mathrm{H}} \mathbf{E}_i \mathbf{V} \mathbf{D}_m|^2 + \sigma^2}, \quad \text{(11a)}$$

$$\text{s.t.} \quad \text{(8b)} - \text{(8c)}, \quad \text{(11b)}$$

where $\tilde{\mu}_k = 1 + \mu_k$ for $k \in \mathcal{K}$. It is emphasized that (P3) and original problem (P1) are equivalence and this transformation ensures no performance loss [15].

Remark 3. *The equivalence between (P1) and (P3) is established as follows: Upon substituting $\boldsymbol{\mu}^*$ back into $f(\mathbf{V},\boldsymbol{\mu})$, the sum-of-logarithms objective function in (P1), i.e., $\sum_{k\in\mathcal{K}} \log_2(1+\gamma_k)$ can be accurately recovered, affirming that (P2) is equivalent to (P1) [40], [41]. In* (9)*, the sole term that depends on $\mathbf{V}$ is $\sum_{k\in\mathcal{K}} \frac{(1+\mu_k)\gamma_k}{1+\gamma_k}$. We can thus eliminate the constant terms related to $\mathbf{V}$ while maintaining the original objective [42]. Hence, the objectives and constraints in (P2) and (P3) are identical, indicating their equivalence. The above equivalences, i.e., (P1) to (P2) and (P2) to (P3), establish the equivalence between the initial problem (P1) and the final version (P3) [40]–[42].*

B. Constrained Optimization on a Manifold

The (P3) solution yields the optimal AP beamforming vectors. However, (P3) has two constraints. Among these, (8c) can be utilized to construct a suitable manifold for solving (P3). Nonetheless, searching solely on this manifold is insufficient because constraint (8b) must also be satisfied. Note that unconstrained searches on manifolds are highly efficient, as demonstrated by recent works [17]–[19]. Consequently, we develop two key steps: identifying the appropriate manifold and devising a penalty method to accommodate constraint (8b). These transform (P3) into an unconstrained search on a manifold, streamlining the optimization process.

*1) **Choosing the right Manifold**:* Usefully, the power constraint is normalized such that the total power meets $\sum_{k\in\mathcal{K}} \|\mathbf{E}_k \mathbf{V} \mathbf{D}_m\|^2 \leq 1$. Each beamforming vector is extended by introducing a modified vector $\tilde{\mathbf{v}}_m = [\tilde{\mathbf{v}}_{m1}^{\mathrm{T}}, \ldots, \tilde{\mathbf{v}}_{mK}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^{(L+1)K\times 1}$, with each $\tilde{\mathbf{v}}_{mk} = [\mathbf{v}_{mk}^{\mathrm{T}}, z_k]^{\mathrm{T}} \in \mathbb{C}^{(L+1)\times 1}$ incorporating an additional element z_k from the auxiliary vector $\mathbf{z} = [z_1, \ldots, z_K]$. This yields the expanded matrix $\tilde{\mathbf{V}} = [\tilde{\mathbf{v}}_1, \ldots, \tilde{\mathbf{v}}_M] \in \mathbb{C}^{(L+1)K\times M}$.

These modifications ensure that the power normalization condition $\sum_{k\in\mathcal{K}} \|\tilde{\mathbf{E}}_k \tilde{\mathbf{V}} \mathbf{D}_m\|^2 = 1$ is satisfied, where $\tilde{\mathbf{E}}_k = [\mathbf{0}, \ldots, \mathbf{I}_{L+1}, \ldots, \mathbf{0}] \in \mathbb{R}^{(L+1)\times(L+1)K}$. A complex oblique manifold is defined as follows:

$$\mathcal{M} = \left\{ \tilde{\mathbf{V}} \in \mathbb{C}^{(L+1)K\times M} \mid \|\tilde{\mathbf{V}}_{:1}\|^2 = \cdots = \|\tilde{\mathbf{V}}_{:M}\|^2 = 1 \right\}. \quad \text{(12)}$$

Consequently, (P3) is transformed into a constrained optimization problem on $\mathcal{M}$, yielding (P4) (13a) (on the top of the next page), where $\hat{\mathbf{h}}_{mk} = \sqrt{p_{\max}}[\mathbf{h}_{mk}, 0]$ and $\hat{\mathbf{a}}(\theta_{mn}) = \sqrt{p_{\max}}[\mathbf{a}(\theta_{mn}), 0]$ adjust the dimensionality and scaling, respectively.

*2) **Dealing with constraint** (13b)**:* In (P4), $\mathcal{M}$ is a Riemannian manifold, where the functions $\hat{f} : \mathcal{M} \to \mathbb{R}$ and $\hat{g}_n : \mathcal{M} \to \mathbb{R}$ are both twice continuously differentiable. Constraint (13b) introduces a significant challenge, as an unconstrained search on $\mathcal{M}$ may not satisfy it. The ALM is employed to address this challenge [15], [43].

The main trick is to add a penalty term to the main objective function. The penalty term punishes the violations of constraints, and the penalty intensity increases gradually. Such adaptations ensure an efficient enforcement of constraints, facilitating the convergence to optimal solutions. Thus, for problem (P4), the set of constraints is given by (13b); thus, whenever $\hat{g}_n(x) > 0$ occurs, a penalty term should be added. There can be several ways to do that. Specifically, the augmented Lagrangian function is defined as [15], [43]

$$\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}) = \hat{f}(\tilde{\mathbf{V}}) + \frac{\rho}{2} \sum_{n\in\mathcal{N}} \max\left\{0, \frac{\lambda_n}{\rho} + \hat{g}_n(\tilde{\mathbf{V}})\right\}^2, \quad \text{(14)}$$

where $\mathcal{N} \triangleq \{1, \ldots, N\}$, $\rho > 0$ serves as a penalty parameter, and $\boldsymbol{\lambda}(\geq 0) \in \mathbb{R}^N$ is the vector of the Lagrange multipliers. For fixed $\boldsymbol{\lambda}$, ρ regulates the severity of the penalty imposed for any violations of (13b) [43]. A high ρ leads to numerical instabilities, whereas too low ρ can fail to penalize constraint violations adequately. Penalty factor ρ is initially set and altered in response to how (13b) is satisfied. If the improvement exceeds a certain level, ρ remains the same; otherwise, it increases to strictly impose (13b). Conversely, $\boldsymbol{\lambda}$ determines the objective function's sensitivity to constraints. They function as a correction term, aiding convergence to the true optimal solution [43]. They are initialized with small

$$\text{(P4)}: \min_{\tilde{\mathbf{V}}\in\mathcal{M}} \hat{f}(\tilde{\mathbf{V}}) = -\sum_{k\in\mathcal{K}} \frac{\tilde{\mu}_k |\sum_{m\in\mathcal{M}_a} \hat{\mathbf{h}}_{mk}^{\mathrm{H}} \tilde{\mathbf{E}}_k \tilde{\mathbf{V}} \mathbf{D}_m|^2}{\sum_{i\in\mathcal{K}} |\sum_{m\in\mathcal{M}_a} \hat{\mathbf{h}}_{mk}^{\mathrm{H}} \tilde{\mathbf{E}}_i \tilde{\mathbf{V}} \mathbf{D}_m|^2 + \sigma^2}, \tag{13a}$$

$$\text{s.t.} \quad \hat{g}_n(\tilde{\mathbf{V}}) = \Gamma_n^{\mathrm{th}} - \sum_{m\in\mathcal{M}_a} \sum_{k\in\mathcal{K}} \hat{\mathbf{a}}^{\mathrm{H}}(\theta_{mn}) \tilde{\mathbf{E}}_k \tilde{\mathbf{V}} \mathbf{D}_m (\tilde{\mathbf{E}}_k \tilde{\mathbf{V}} \mathbf{D}_m)^{\mathrm{H}} \hat{\mathbf{a}}(\theta_{mn}) \leq 0, \ \forall n \tag{13b}$$

positive values and iterated to enforce constraints to gradually balance the constraints and the objective.

The ALM employs a cyclical strategy where $\tilde{\mathbf{V}}$ is optimized with fixed $\boldsymbol{\lambda}$, an unconstrained search on $\mathcal{M}$, followed by an update of $\boldsymbol{\lambda}$ [43]. This approach is called ALMO [13]. Thus, problem (P4) is recast as

$$\text{(P5)}: \min_{\tilde{\mathbf{V}}\in\mathcal{M},\boldsymbol{\lambda}} \quad \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}). \tag{15a}$$

Next, the detailed steps to solve (P5) are outlined.

*3) **The proposed ALMCI algorithm**:* This minimizes smooth function $\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$ over complex oblique manifold $\mathcal{M} \to \mathbb{R}$.

The algorithm computes the Riemannian gradient, $\mathrm{grad}_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$, at current point $\tilde{\mathbf{V}}_t$ [13], [15]. Then, the next point, $\tilde{\mathbf{V}}_{t+1}$, is obtained via a retraction mapping $\mathcal{R}_{\tilde{\mathbf{V}}_t}$, projecting search direction vector $\eta\tilde{\mathbf{V}}_t$ back onto $\mathcal{M}$. A vector transport function $\mathcal{T}_{\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}_{t+1}}$ then transfers the conjugate search direction from $T_{\tilde{\mathbf{V}}_t}\mathcal{M}$ to $T_{\tilde{\mathbf{V}}_{t+1}}\mathcal{M}$, ensuring the manifold's curvature and geometric properties. The reader is referred to Fig. 2 in [15] for a visual representation of the MO steps. The main steps are summarized as follows:

(a) **Riemannian gradient**: This is obtained by orthogonally projecting Euclidean gradient $\nabla_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$ onto tangent space $T_{\tilde{\mathbf{V}}_t}\mathcal{M}$. At $\tilde{\mathbf{V}}_t$, it can be obtained as

$$\begin{aligned}\mathrm{grad}_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}) &= \nabla_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}) \\ &\quad - \Re\{\nabla_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}) \circ \tilde{\mathbf{V}}_t^*\} \circ \tilde{\mathbf{V}}_t,\end{aligned} \tag{16}$$

where

$$T_{\tilde{\mathbf{V}}_t}\mathcal{M} = \left\{ \mathbf{c} \in \mathbb{C}^{(L+1)K} \mid \Re\{\mathbf{c} \circ \mathbf{V}_t^*\} = \mathbf{0}_{(L+1)K} \right\}, \tag{17}$$

and $\mathbf{c} \in \mathbb{C}^{(L+1)K}$. The Euclidean gradient term in (16) is derived by differentiating (14) and is given in (18).

(b) **Search direction and mapping**: By extending traditional EO techniques using $\mathrm{grad}_{\tilde{\mathbf{V}}_t} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$, the update rule for the search direction is given by

$$\boldsymbol{\eta}_{t+1} = -\mathrm{grad}_{\tilde{\mathbf{V}}_{t+1}} \mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda}) + \beta_t \mathcal{T}_{\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}_{t+1}}(\boldsymbol{\eta}_t), \tag{19}$$

where $\boldsymbol{\eta}_t$ is the current search direction and β_t is computed using the Hestenes-Stiefel approach [44]. Here, $\mathcal{T}_{\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}_{t+1}}(\boldsymbol{\eta}_t)$ is mapping a vector from the tangent space at $\tilde{\mathbf{V}}_t$, $T_{\tilde{\mathbf{V}}_t}\mathcal{M}$, to the tangent space at $\tilde{\mathbf{V}}_{t+1}$, $T_{\tilde{\mathbf{V}}_{t+1}}\mathcal{M}$, which is given by

$$\mathcal{T}_{\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}_{t+1}}(\boldsymbol{\eta}_t) = \boldsymbol{\eta}_t - \Re\{\boldsymbol{\eta}_t \circ \tilde{\mathbf{V}}_{t+1}^*\} \circ \tilde{\mathbf{V}}_{t+1}. \tag{20}$$

Note that since $\boldsymbol{\eta}_t$ and $\boldsymbol{\eta}_{t+1}$ belong to different tangent spaces, thereby a vector transport is needed to maintain the integrity of the manifold's geometry.

(c) **Retraction**: Once search direction $\boldsymbol{\eta}_t$ is established at $\tilde{\mathbf{V}}_t$, a retraction is employed to smoothly map this direction back onto the manifold, determining the next point $\tilde{\mathbf{V}}_{t+1}$. This retraction process is defined as

$$\tilde{\mathbf{V}}_{t+1} = \mathcal{R}_{\tilde{\mathbf{V}}_t}(\alpha_t \boldsymbol{\eta}_t) = \mathrm{unt}(\alpha_t \boldsymbol{\eta}_t), \tag{21}$$

where α_t represents the step size along direction $\boldsymbol{\eta}_t$. Thus, MO proceeds iteratively to converge to an optimal point where the Riemannian gradient vanishes, as defined in (14).

(d) **Updating the Lagrange multipliers**: Once $\tilde{\mathbf{V}}$ is optimized on $\mathcal{M}$, $\boldsymbol{\lambda}$ is updated to meet the constraints. These updates include clipping safeguards that bound the multipliers, ensuring that updates effectively contribute to resolving constraint violations. The update rule for each Lagrange multiplier at iteration t is specified by

$$\lambda_n^{t+1} = \mathrm{clip}_{[\lambda_n^{\min}, \lambda_n^{\max}]} \left(\lambda_n^t + \rho_t \hat{g}_n(\tilde{\mathbf{V}}_{t+1}) \right), \ \forall n, \tag{22}$$

where $\rho_t > 0$ is the positive penalty parameter at iteration t. The clipping process restricts each λ_n^{t+1} to a predetermined range $[\lambda_n^{\min}, \lambda_n^{\max}]$, preventing unbounded growth and maintaining stability [13], [15]. Moreover, the updates occur only when significant progress is made in reducing constraint violations, thus avoiding premature or overly aggressive adjustments that may cause instability.

The overall process is given in Algorithm 1. More specifically, at each iteration, the maximum constraint violation, $\max_n |\sigma_n|$ is evaluated, where σ_n represents the adjusted constraint residual from step 19 of Algorithm 1. If the maximum violation decreases significantly compared to the previous iteration, the penalty parameter ρ remains constant in step 22, as the constraints are adequately enforced. Otherwise, if the violation does not decrease sufficiently, the penalty parameter is increased multiplicatively in step 24, with $\theta_\rho > 1$ as the penalty increment factor.

Fig. 2 depicts the evaluation of a cost function of problem (P5), where parameters t_1 and t_2 are varied across a two-dimensional grid spanning the interval $[-1, 1]$. In addition, $\mathbf{d}_1$ and $\mathbf{d}_2$ are tangent vectors at point $\tilde{\mathbf{V}}$ on manifold $\mathcal{M}$. These vectors represent directions in which we can move from $\tilde{\mathbf{V}}$ while remaining on the manifold. They are obtained by orthogonally projecting the Euclidean gradient onto the tangent space at $\tilde{\mathbf{V}}$. This figure captures the concavity of augmented Lagrangian cost function ($\mathcal{L}_\rho$) evaluated at the point $\mathcal{R}_{\tilde{\mathbf{V}}}(\alpha_1\eta_1 + \alpha_2\eta_2)$ where $\mathcal{R}$ denotes the Retraction map on the manifold $\mathcal{M}$. It highlights how the cost changes in response to movements along the manifold in the directions specified by these tangent vectors. This also serves as a visual guide to the algorithm's progress toward convergence, validating that each iteration moves closer to the solution by reducing the cost function value.

Note that our problem formulation and solution are easily adaptable to additional constraints on (P1), such as minimum user rates, making it more versatile.

$$\nabla_{\tilde{\mathbf{V}}_t}\mathcal{L}_\rho(\tilde{\mathbf{V}},\boldsymbol{\lambda}) = \sum\nolimits_{k\in\mathcal{K}} -\tilde{\mu}_k \Bigg(\frac{2\sum_{m\in\mathcal{M}_a}\hat{\mathbf{h}}_{mk}^{\mathrm{H}}\tilde{\mathbf{E}}_k\tilde{\mathbf{V}}_t\mathbf{D}_m\sum_{m\in\mathcal{M}_a}\tilde{\mathbf{E}}_k^{\mathrm{H}}\hat{\mathbf{h}}_{mk}\mathbf{D}_m^{\mathrm{H}}}{\sum_{j\in\mathcal{K}}|\sum_{m\in\mathcal{M}_a}\hat{\mathbf{h}}_{mk}^{\mathrm{H}}\tilde{\mathbf{E}}_j\tilde{\mathbf{V}}\mathbf{D}_m|^2+\sigma_k^2}$$
$$- \sum\nolimits_{i\in\mathcal{K}} \frac{2|\sum_{m\in\mathcal{M}_a}\hat{\mathbf{h}}_{mk}^{\mathrm{H}}\tilde{\mathbf{E}}_k\tilde{\mathbf{V}}\mathbf{D}_m|^2\sum_{m\in\mathcal{M}_a}\hat{\mathbf{h}}_{mk}^{\mathrm{H}}\tilde{\mathbf{E}}_i\tilde{\mathbf{V}}_t\mathbf{D}_m\sum_{m\in\mathcal{M}_a}\tilde{\mathbf{E}}_i^{\mathrm{H}}\hat{\mathbf{h}}_{mk}\mathbf{D}_m^{\mathrm{H}}}{\left(\sum_{j\in\mathcal{K}}|\sum_{m\in\mathcal{M}_a}\hat{\mathbf{h}}_{mk}^{\mathrm{H}}\tilde{\mathbf{E}}_j\tilde{\mathbf{V}}\mathbf{D}_m|^2+\sigma_k^2\right)^2}\Bigg)$$
$$- 2\rho\sum\nolimits_{n\in\mathcal{N}}\mathbf{1}_{\left\{\lambda_n+\frac{\hat{g}_n(\tilde{\mathbf{V}})}{\rho}\right\}}\left(\frac{\lambda_n}{\rho}+\hat{g}_n(\tilde{\mathbf{V}})\right)\sum\nolimits_{m\in\mathcal{M}_a}\sum\nolimits_{k\in\mathcal{K}}\tilde{\mathbf{E}}_k^{\mathrm{H}}\hat{\mathbf{a}}(\theta_{mn})\left(\hat{\mathbf{a}}^{\mathrm{H}}(\theta_{mn})\tilde{\mathbf{E}}_k\tilde{\mathbf{V}}_t\mathbf{D}_m\mathbf{D}_m^{\mathrm{H}}\right) \quad (18)$$

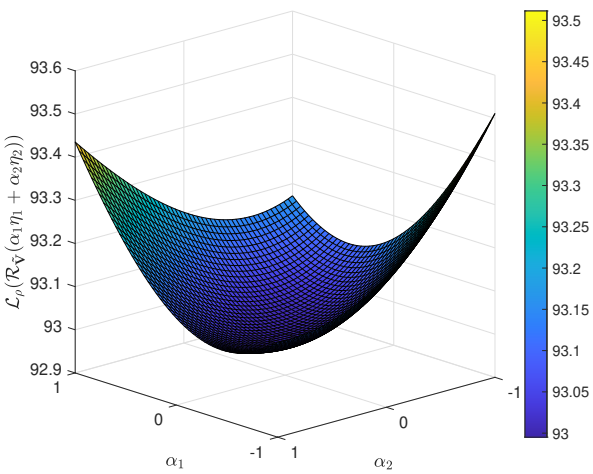


Fig. 2: The cost function of (P5).

The computational complexity for this process is dominated by the iterations of Algorithm 1, yielding an approximate complexity of $\mathcal{O}(T(LMK+LMK^3))$, where T denotes the total iterations.

C. CCPA

This method is based on standard SDR and SCA techniques. It is equivalent to algorithms in [2]–[4] and is developed as a comparative benchmark.

By defining $\mathbf{W}_k=\mathbf{w}_k\mathbf{w}_k^{\mathrm{H}}$ for $k\in\mathcal{K}$, where $\mathbf{w}_k\in\mathbb{C}^{LM\times 1}$ is the beamformers of all APs, and $\mathbf{W}_k\in\mathbb{C}^{LM\times LM}$ is a semi-definite matrix, i.e., $\mathbf{W}_k\succeq 0$, with $\mathrm{Rank}(\mathbf{W}_k)=1$, (P1) is transformed into a standard semi-definite problem (SDP) by relaxing the rank one constraint, as outlined in (P6) (23) [4]. Additionally, the channel matrices for each AP to users and sensing targets are redefined as $\mathbf{f}_k=[\mathbf{h}_{1k}^{\mathrm{T}},\ldots,\mathbf{h}_{Mk}^{\mathrm{T}}]^{\mathrm{T}}\in\mathbb{C}^{LM\times 1}$ and $\mathbf{g}_n=[\mathbf{a}^{\mathrm{T}}(\theta_{1n}),\ldots,\mathbf{a}^{\mathrm{T}}(\theta_{Mn})]^{\mathrm{T}}\in\mathbb{C}^{LM\times 1}$, respectively, and the matrix $\mathbf{D}_m'\in\mathbb{R}^{LM\times LM}$ is a block diagonal matrix, composed of M blocks, each corresponding to a AP. This configuration is represented by matrix $\mathrm{diag}(\mathbf{0},\ldots,\mathbf{0},\mathbf{I}_L,\mathbf{0},\ldots,\mathbf{0})$, where $\mathbf{I}_L$ is placed at the m-th block on the diagonal. This placement specifically activates the transmit antennas of the m-th AP. Note that the interior-point method can solve sub-problems using the SDP [42]. Finally, to impose the relaxed rank one constraint, Gaussian randomization is utilized [45].

According to [46, Th. 3.12], the order of complexity for a SDP problem with m SDP constraints, which includes a $n\times n$ positive semi-definite matrix is given by $\mathcal{O}\left(\sqrt{n}\log\left(\frac{1}{\epsilon}\right)(mn^3+m^2n^2+m^3)\right)$ with $\epsilon>0$. For problem (23), with $n=LM$ and $m=N+K+1$, the approximate computational complexity for solving (23) is $\mathcal{O}\left(\sqrt{LM}\log\left(\frac{1}{\epsilon}\right)\left((N+K)(LM)^3\right)\right)$.

D. MCQT-SCA

This method introduces auxiliary variables to simplify and solve problem (P1). Specifically, (P1) is first converted to (P2) given in (9) with the optimal value of $\boldsymbol{\mu}$ based on (10). However, (P2) remains challenging due to its high-dimensional sum-of-fraction non-convex objective function and the sensing constraint. As it is not applicable to common FP methods, such as Dinkelbach's algorithm, we employ the MCQT [41]. MCQT extends scalar-form fractional programming to matrix-form, handling the non-convexity of high-dimensional fractions [41]. By introducing another set of auxiliary variables $\boldsymbol{\zeta}=[\zeta_1,\ldots,\zeta_K]^{\mathrm{T}}\in\mathbb{C}^{K\times 1}$, (P2) is reformulated into (P7) given in (24), where

$$g\left(\mathbf{V},\mathbf{a}(\theta_{mn})\right)=\sum_{m\in\mathcal{M}_a}\sum_{k\in\mathcal{K}}\mathbf{a}^{\mathrm{H}}(\theta_{mn})\mathbf{S}_{mk}\mathbf{V}(\mathbf{S}_{mk}\mathbf{V})^{\mathrm{H}}\mathbf{a}(\theta_{mn}). \quad (25)$$

In addition, $\mathbf{A}=\mathbf{I}_K\otimes\sum_{k\in\mathcal{K}}\mathbf{f}_k\zeta_k\zeta_k^{\mathrm{H}}\mathbf{f}_k^{\mathrm{H}}$, $\mathbf{B}=\left[\mathbf{b}_1^{\mathrm{T}},\ldots,\mathbf{b}_K^{\mathrm{T}}\right]^{\mathrm{T}}$ with $\mathbf{b}_k^{\mathrm{T}}=\sqrt{\tilde{\mu}_k}\mathbf{f}_k\zeta_k$, and $\mathbf{D}_m''=\mathbf{I}_K\otimes\{\mathbf{e}_m\mathbf{e}_m^{\mathrm{H}}\otimes\mathbf{I}_L\}$ with $\mathbf{e}_m\in\mathbb{R}^{M\times 1}$. In (24b), selection matrix $\mathbf{S}_{mk}$ extracts $\mathbf{v}_{mk}$ from $\mathbf{V}$, and defined as $\mathbf{S}_{mk}=[\mathbf{0},\cdots,\mathbf{0},\mathbf{I}_L,\mathbf{0},\cdots,\mathbf{0}]\in\mathbb{R}^{L\times KLM}$, where identity matrix $\mathbf{I}_L$ starts at column index $(m-1)KL+(k-1)L+1$ and spans L columns. This places identity matrix $\mathbf{I}_L$ at the block corresponding to the position of $\mathbf{v}_{mk}$ within vector $\mathbf{V}$. Moreover, the constraint in (24b) is linearized using the SCA method [5]. Therefore, this baseline is referred to as MCQT-SCA. The algorithm iteratively updates $\boldsymbol{\zeta}$ and $\mathbf{V}$. Optimal $\boldsymbol{\zeta}$ can be obtained by

$$\zeta_k^*=\frac{\sqrt{\mu_k}\mathbf{f}_k^{\mathrm{H}}\mathbf{v}_k}{\sum_{j\in\mathcal{K}}\mathbf{f}_k^{\mathrm{H}}\mathbf{v}_j\left(\mathbf{f}_k^{\mathrm{H}}\mathbf{v}_j\right)^{\mathrm{H}}+\sigma^2},\ \forall k, \quad (26)$$

where $\mathbf{v}_k=[\mathbf{v}_{mk}^{\mathrm{T}},\ldots,\mathbf{v}_{Mk}^{\mathrm{T}}]^{\mathrm{T}}\in\mathbb{C}^{LM\times 1}$. Problem (P7) is a standard quadratically constrained quadratic program (QCQP) and can be efficiently solved by the interior-point method [42]. In CF networks, the channels are high-dimensional, making matrix $\mathbf{A}$ considerably large (MLK). Thus, inverting $\mathbf{A}$ has a complexity of about $\mathcal{O}(M^3L^3K^3)$. This significant computational burden may impede the practical implementation of the precoding design using this method.

For the MCQT-SCA and CCPA algorithms, a feasibility problem is solved before solving problems (P6) and (P7). This step generates an initial point that satisfies all constraints and serves as the starting point for the iterative algorithms.

$$\text{(P6)}: \max_{\{\mathbf{W}_k\}_{k\in\mathcal{K}}} \sum\nolimits_{k\in\mathcal{K}} \log_2\left(\sum\nolimits_{i\in\mathcal{K}} \mathbf{f}_k^{\mathrm{H}}\mathbf{W}_i\mathbf{f}_k+\sigma^2\right) - \log_2\left(\sum\nolimits_{i\in\mathcal{K}_k} \mathbf{f}_k^{\mathrm{H}}\mathbf{W}_i^{(t)}\mathbf{f}_k+\sigma^2\right) + \frac{\log_2(e)}{2^{\log_2\left(\sum_{i\in\mathcal{K}_k}\mathbf{f}_k^{\mathrm{H}}\mathbf{W}_i^{(t)}\mathbf{f}_k+\sigma^2\right)}} \sum\nolimits_{i\in\mathcal{K}_k} \mathbf{f}_k^{\mathrm{H}}\left(\mathbf{W}_i-\mathbf{W}_i^{(t)}\right)\mathbf{f}_k, \quad (23a)$$

$$\text{s.t.} \quad \mathrm{Tr}\left(\sum\nolimits_{m\in\mathcal{M}_a}\sum\nolimits_{k\in\mathcal{K}} \mathbf{D}'_m\mathbf{g}_n\mathbf{g}_n^{\mathrm{H}}\mathbf{D}'_m\mathbf{W}_k\right) \geq \Gamma_n^{\mathrm{th}}, \ \forall n, \quad (23b)$$

$$\mathrm{Tr}\left(\sum\nolimits_{k\in\mathcal{K}}\mathbf{W}_k\right) \leq p_{\max}, \quad (23c)$$

$$\mathbf{W}_k \succeq 0, \ \forall k \quad (23d)$$

$$\text{(P7)}: \min_{\mathbf{V},\{\zeta_k\}_{k\in\mathcal{K}},\{\mu_k\}_{k\in\mathcal{K}}} \mathbf{V}^{\mathrm{H}}\mathbf{A}\mathbf{V} - \Re\{2\mathbf{B}^{\mathrm{H}}\mathbf{V}\}, \quad (24a)$$

$$\text{s.t.} \quad g\left(\mathbf{V}^{(t)}, \mathbf{a}(\theta_{mn})\right) + 2\mathbf{S}_{mk}^{\mathrm{T}}\mathbf{a}(\theta_{mn})\left(\mathbf{a}(\theta_{mn})^{\mathrm{H}}\mathbf{S}_{mk}\mathbf{V}^{(t)}\right)\left(\mathbf{V}-\mathbf{V}^{(t)}\right) \geq \Gamma_n^{\mathrm{th}}, \ \forall n, \quad (24b)$$

$$\mathbf{V}^{\mathrm{H}}\mathbf{D}''_m\mathbf{V} \leq p_{\max}, \ \forall m \quad (24c)$$

While our proposed algorithm is more efficient than benchmarks, its K^3 complexity may pose challenges for a high number of users. Nevertheless, it significantly improves computational efficiency over state-of-the-art methods (e.g., Fig. 4), making it a viable low-complexity alternative in practical scenarios. Several strategies can be employed in future research to address potential challenges with a high number of users. These include (i) distributed or parallel processing, which divides the optimization problem across multiple cores or nodes, thereby significantly reducing the overall computational burden, and (ii) user/target grouping or clustering, where users or targets with similar characteristics or located within the same geographical area are aggregated into representative groups, enabling optimization to be performed at the group level and reducing the problem's dimensionality.

V. Convergence and Optimality

Next, we analyze the convergence conditions of the ALMCI algorithm for (P5). Specifically, we assess whether it converges to a global minimizer of (P5), assuming that Algorithm 1 operates with global accuracy

Proposition 1. *Consider executing the ALMCI algorithm with $\epsilon_{\min}=0$, producing an infinite sequence $\{\epsilon_t\}$ that converges to zero. During each iteration t, Algorithm 1 determines a candidate solution $\tilde{\mathbf{V}}_{t+1}$ ensuring it fulfills the following criterion:*

$$\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1},\boldsymbol{\lambda}_t) \leq \mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t)+\epsilon_t, \quad (27)$$

where $\tilde{\mathbf{V}}_{t+1}$ is a feasible global minimizer of (P5). If the sequence $\{\tilde{\mathbf{V}}_t\}_{t=0}^{\infty}$ produced by the ALMCI algorithm, converges to a limit point $\tilde{\mathbf{V}}^*$, then $\tilde{\mathbf{V}}^*$ serves as a global minimizer of (P5) as well.

Proof. The convergence to a global minimizer can be justified by leveraging the findings of Theorem 1 and Theorem 2 in [47], which confirms that any limit point of the sequence produced by the ALMCI algorithm meets the global optimality conditions for (P5).

1) Boundedness and non-increasing sequences: Given the condition:

$$\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1},\boldsymbol{\lambda}_t) \leq \mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t)+\epsilon_t, \quad (28)$$

and the fact that $\epsilon_t \to 0$, we have:

$$\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1},\boldsymbol{\lambda}_t) \leq \mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t). \quad (29)$$

This shows that the sequence $\{\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t)\}$ is monotonically non-increasing and is bounded below by the global minimum of the Lagrangian function.

2) Convergence to a limit point: Consider the sequence $\{\tilde{\mathbf{V}}_t\}_{t=1}^{T}$ has a limit point $\tilde{\mathbf{V}}^*$. According to the Bolzano-Weierstrass theorem, any bounded sequence in $\mathbb{R}^T$ contains a convergent subsequence [48]. Therefore, a convergent subsequence $\{\tilde{\mathbf{V}}_{t_k}\}_{k=1}^{T}$ exists, satisfying $\{\tilde{\mathbf{V}}_{t_k}\}_{k=1}^{T} \to \tilde{\mathbf{V}}^*$.

3) Feasibility properties : To verify that $\tilde{\mathbf{V}}^*$ satisfies the feasibility conditions of the original problem (P5), consider the following reasoning. The augmented Lagrangian function $\mathcal{L}_\rho(\mathbf{V},\boldsymbol{\lambda})$ incorporates penalty terms designed to penalize constraint violations. Hence, the minimizer of $\mathcal{L}_\rho(\mathbf{V},\boldsymbol{\lambda})$ ensures that these violations diminish over iterations. In particular, the sequence $\{\tilde{\mathbf{V}}_t\}_{t=1}^{T}$ is generated such that $\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t)$ is monotonically decreasing and convergent. Consequently, the penalty terms approach zero, otherwise, the augmented Lagrangian grows unbounded. Hence, it follows that:

$$\lim_{t\to\infty}\sum_n \max\left\{0, \frac{\lambda_n^t}{\rho_t}+\hat{g}_n(\tilde{\mathbf{V}}_{t+1})\right\}=0,$$

which shows that $\hat{g}_n(\tilde{\mathbf{V}}^*) \leq 0$, proving that $\tilde{\mathbf{V}}^*$ is feasible for (P5).

4) Global optimality analysis: To prove the global optimality of $\tilde{\mathbf{V}}^*$, it must be shown that $f(\tilde{\mathbf{V}}^*) \leq f(\mathbf{V})$ for all feasible $\mathbf{V}$. By leveraging (27) and recognizing that $\epsilon_t \to 0$, we can conclude:

$$\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1},\boldsymbol{\lambda}_t) \leq \mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_t,\boldsymbol{\lambda}_t). \quad (30)$$

As $\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}^*$, it follows that $\tilde{\mathbf{V}}_{t+1} \to \tilde{\mathbf{V}}^*$. Let $\mathbf{V}$ be any feasible point for (P5). For sufficiently large t, since $\tilde{\mathbf{V}}_t \to \tilde{\mathbf{V}}^*$ and the penalty terms vanish at feasibility, we have:

$$\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}^*,\boldsymbol{\lambda}_t) \approx f(\tilde{\mathbf{V}}^*) \quad \text{and} \quad \mathcal{L}_{\rho_t}(\mathbf{V},\boldsymbol{\lambda}_t) \approx f(\mathbf{V}). \quad (31)$$

Therefore, $f(\tilde{\mathbf{V}}^*) \leq f(\mathbf{V})$ for every feasible $\mathbf{V}$. Consequently, the limit point $\tilde{\mathbf{V}}^*$ of the sequence $\{\tilde{\mathbf{V}}_t\}_{t=1}^{T}$, generated by the ALMCI algorithm, is a global minimizer of (P5). This concludes the proof of Proposition 1. □

Algorithm 1 ALMCI Algorithm

Require: $\mathcal{M}$, $\hat{f}(\tilde{\mathbf{V}})$, $\{\hat{g}_n(\tilde{\mathbf{V}})\}_{n\in\mathcal{N}}$: $\mathcal{M} \to \mathbb{R}$.
Initialization: Initial point $\tilde{\mathbf{V}}_0 \in \mathcal{M}$, Lagrange multipliers $\boldsymbol{\lambda}^0 \in \mathbb{R}^N$, accuracy tolerance $\epsilon_{\min}$, convergence tolerance $\delta_1 > 0$, initial accuracy $\epsilon_0 > 0$, initial penalty factor ρ_0, reduction factors $\theta_\epsilon \in (0,1)$ and $\theta_\rho > 1$, boundaries for the multipliers $\lambda_n^{\min}, \lambda_n^{\max} \in \mathbb{R}$ ensuring $\lambda_n^{\min} \leq \lambda_n^{\max}$, ratio $\tau \in (0,1)$, a minimum acceptable distance $d_{\min}$, and final convergence tolerance $\delta_2 > 0$.
Set $t = 0$.
while $\text{dist}(\hat{f}(\tilde{\mathbf{V}}_t), \hat{f}(\tilde{\mathbf{V}}_{t+1})) \geq \delta_2$ **do**
  Set $j = 0$.
  while $\text{dist}(\tilde{\mathbf{V}}_j, \tilde{\mathbf{V}}_{j+1}) \geq d_{\min}$ or $\epsilon_j > \epsilon_{\min}$ **do**
    Set $i = 0$.
    Calculate $\boldsymbol{\eta}_0 = -\text{grad}_{\tilde{\mathbf{V}}_0}\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$ as Riemannian gradient according to (16).
    while $\|\text{grad}_{\tilde{\mathbf{V}}_i}\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})\|_2 > \delta_1$ **do**
      Determine the Armijo backtracking line search step size α_i according to [44].
      Update $\tilde{\mathbf{V}}_{i+1}$ using retraction $\mathcal{R}_{\tilde{\mathbf{V}}_i}(\alpha_i \boldsymbol{\eta}_i)$ as described in (21).
      Compute the Riemannian gradient at the new point $\text{grad}_{\tilde{\mathbf{V}}_{i+1}}\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$ according to (16).
      Obtain the vector transport $\mathcal{T}_{\tilde{\mathbf{V}}_i \to \tilde{\mathbf{V}}_{i+1}}(\boldsymbol{\eta}_i)$ using (20).
      Compute the Hestenes-Stiefel parameter β_i according to [44].
      Update the conjugate gradient direction based on (19).
      $i \leftarrow i + 1$;
    end while
    Update the Lagrange multiplier using (22).
    Set $\sigma_n^{j+1} = \max\left\{\hat{g}_n(\tilde{\mathbf{V}}_{j+1}), -\frac{\lambda_n^{j+1}}{\rho_t}\right\}, \forall n$.
    Adjust the accuracy tolerance $\epsilon_{j+1} = \max\{\epsilon_{\min}, \theta_\epsilon \epsilon_j\}$.
    if $j = 0$ or $\max_n\{|\sigma_n^{j+1}|\} \leq \tau \max_n\{|\sigma_n^j|\}$ **then**
      Maintain the current penalty value: $\rho_{j+1} = \rho_j$.
    else
      Increment the penalty value: $\rho_{j+1} = \theta_\rho \rho_j$.
    end if
    $j \leftarrow j + 1$
  end while
  Calculate γ according to the update rule defined in (10).
  Obtain $\tilde{\mathbf{V}}^{(t+1)}$.
  $t \leftarrow t + 1$
end while
Output: $\mathbf{V}^* = \tilde{\mathbf{V}}^*(1 : LK, M)$.

Although attaining global optimality can be difficult, convergence to stationary points is generally achievable. The following proposition outlines the case for first-order convergence.

Proposition 2. *Consider the ALMCI algorithm with $\epsilon_{\min} = 0$. Assume that at each iteration t, Algorithm 1 generates a point $\tilde{\mathbf{V}}_{t+1}$ that satisfies:*

$$\left\|\text{grad}_{\tilde{\mathbf{V}}}\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1}, \boldsymbol{\lambda}_t)\right\| \leq \epsilon_t. \tag{32}$$

If the sequence $\{\tilde{\mathbf{V}}_t\}_{t=0}^{\infty}$ converges to a limit point $\tilde{\mathbf{V}}^*$ within the feasible region of (P2), then $\tilde{\mathbf{V}}^*$ satisfies the Karush-Kuhn-Tucker (KKT) conditions of (P5) [13, Proposition 3.2].

Proof. At the initial iteration ($t = 0$), we start with a feasible point $\tilde{\mathbf{V}}_0 \in \mathcal{M}$ and a set of Lagrange multipliers $\boldsymbol{\lambda}^0 \in \mathbb{R}^N$. Assuming the algorithm converges by iteration t, it implies that $\tilde{\mathbf{V}}_t$ is a stationary point of the Lagrangian function, associated with the current multipliers $\boldsymbol{\lambda}^t$ and penalty parameters $\{\rho_t\}$. Given that $\epsilon_{\min} = 0$ and $\epsilon_t \to 0$ as $t \to \infty$, it follows that:

$$\left\|\text{grad}_{\tilde{\mathbf{V}}}\mathcal{L}_{\rho_t}(\tilde{\mathbf{V}}_{t+1}, \boldsymbol{\lambda}_t)\right\| \to 0 \text{ as } t \to \infty. \tag{33}$$

By the continuity of the gradient operator and the Lagrangian function, we have $\text{grad}_{\tilde{\mathbf{V}}}\mathcal{L}_{\rho^*}(\tilde{\mathbf{V}}^*, \boldsymbol{\lambda}^*) = 0$, where $\rho^* = \lim_{t\to\infty} \rho_t$ and $\boldsymbol{\lambda}^* = \lim_{t\to\infty} \boldsymbol{\lambda}_t$. Since $\tilde{\mathbf{V}}^*$ lies within the feasible set of (P2), it satisfies the primal feasibility conditions. The dual feasibility and complementary slackness conditions are satisfied due to the properties of the Lagrange multipliers $\boldsymbol{\lambda}^*$ [13]. Thus, $\tilde{\mathbf{V}}^*$ satisfies the KKT conditions of (P5), completing the proof.

Considering the $t-1$ iteration, the update of the Lagrange multipliers via (22) ensures that $\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$ remains bounded below, as the penalty terms associated with constraint violations are always non-negative. The multipliers are also constrained within the bounds $\lambda_n^{\min} \leq \lambda_n^{\max}$. Given the properties of the Riemannian gradient and the compactness of the manifold $\mathcal{M}$, the sequence $\{\tilde{\mathbf{V}}_t\}_{t=0}^{\infty}$ is confined to a compact subset of $\mathcal{M}$, implying the existence of a convergent subsequence. Let $\tilde{\mathbf{V}}^*$ denote the limit of this subsequence. By the continuity of $\hat{f}(\tilde{\mathbf{V}})$ and $\hat{g}_n(\tilde{\mathbf{V}})$, $\tilde{\mathbf{V}}^*$ is a stationary point of $\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$. Moreover, the algorithm's rule for adjusting the penalty parameters $\{\rho_t\}$ ensures that, in the limit, constraint violations are sufficiently penalized. This guarantees that any limit point of the sequence $\{\tilde{\mathbf{V}}_t\}_{t=0}^{\infty}$ must satisfy the constraints, making $\tilde{\mathbf{V}}^*$ a feasible solution to the original problem. Thus, by applying the first-order optimality conditions on Riemannian manifolds and the properties of the ALM, we conclude that the sequence $\{\tilde{\mathbf{V}}_t\}_{t=0}^{\infty}$ converges to a stationary point of $\mathcal{L}_\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})$. □

TABLE I: Simulation and algorithm parameters.

Parameter	Value	Parameter	Value
σ^2	$-80\,\text{dBm}$	Γ_n^{th}	$20\,\text{dBm}$
L	16	$p_{\max}$	$30\,\text{dBm}$
K	2	δ_1, δ_2	10^{-6}
N	4	ν	2
$d_{\min}$	10^{-10}	ϵ_0	10^{-3}
$\epsilon_{\min}$	10^{-6}	τ	0.5
θ_ϵ	0.5	θ_ρ	0.25
ρ_0	1	$\{\lambda_n^{\min}, \lambda_n^{\max}\}$	{0,100}

VI. Simulation Results

This section presents the simulation results, demonstrating the performance of the proposed optimal transmit beamforming strategy. Detailed comparisons with benchmark methods demonstrate the benefits of the proposed algorithm.

A. Simulation Setup and Parameters

Herein, the configuration and parameters of the simulation are detailed. Unless specified otherwise, Table I lists the simulation parameters [8]–[10], [22]. The path-loss is modeled as $\zeta_p = C_0 \left(\frac{d_p}{D_0}\right)^{-\nu}$, where $\zeta = \{\zeta_{\mathbf{h}_{mk}}, \zeta_{mn}\}$, $C_0 = -30\,\mathrm{dB}$ represents the path-loss at the reference distance $D_0 = 1\,\mathrm{m}$, $d_p = \{d_{mk}, d_{mn}\}$ represents the individual link distance, and ν denotes the path-loss exponent.

A system with two APs is simulated. Users and sensing targets are uniformly distributed within a $500 \times 500\,\mathrm{m}^2$ area. Each simulation result is averaged over 10^3 Monte Carlo trials.

To study beampattern gains, AP-1 and AP-2 are placed at coordinates $(10, 10)$ and $(80, 80)$, respectively. From AP-1, the direction angles for users and sensing targets are $\{-40, 40\}^\circ$ and $\{-80, -20, 20, 80\}^\circ$, respectively, ensuring coverage of a broad angular range. Similarly, from AP-2, direction angles for users and sensing targets are $\{-50, 50\}^\circ$ and $\{-70, -10, 10, 70\}^\circ$, respectively. The circles and triangles denote communication users and sensing targets, respectively.

B. Benchmark Schemes

The following four benchmarks are used to evaluate the proposed MO approach comparatively. The first two, MMSE and zero-forcing (ZF) beamformers, are known for their computational efficiency and ability to reduce interference [28]. We derived the last two based on previous works [2]–[5], which do not use the MO approach.

1) **Sensing-ignorant ZF beamforming:** This beamformer is given by $\mathbf{V}_{\mathrm{ZF},m} = \mathbf{H}_m(\mathbf{H}_m^{\mathrm{H}}\mathbf{H}_m)^{-1}$, where $\mathbf{H}_m = [\mathbf{h}_{m1}, \ldots, \mathbf{h}_{mK}] \in \mathbb{C}^{L\times K}$ for $k \in \mathcal{K}$ and $m \in \mathcal{M}_a$ is the channel matrix from the m-th AP to all user. This only focuses solely on the communication users and eliminates interference from non-intended users.
2) **Sensing-ignorant MMSE beamforming:** This beamformer can be represented as $\mathbf{V}_{\mathrm{MMSE},m} = \mathbf{H}_m\left(\mathbf{H}_m^{\mathrm{H}}\mathbf{H}_m + \sigma^2\mathbf{I}_K\right)^{-1}$ for $m \in \mathcal{M}_a$. This is also designed for communication only. This minimizes the error between the estimated and actual signals and effectively balances the signal and noise.
3) **CCPA:** This is equivalent to the joint communication and sensing beamforming algorithm developed in [4]. It solves (P1) using conventional SDR and SCA techniques (see Section IV-C). Thus, this benchmark allows our algorithm to be compared against the state-of-the-art.
4) **MCQT-SCA:** This evaluates the beamforming method from [5]. It handles (P1) with the MCQT and SCA approaches (see Section IV-D) [41].

C. Convergence and Execution Times

Table II compares the convergence of ALMCI, CCPA, and MCQT-SCA for different simulation setups. At $p_{\max} = 25\,\mathrm{dBm}$ with $L = 8$ antennas, ALMCI stabilizes at 26.7211 (bps/Hz) sum rate by the fourth iteration. In contrast, the other two converge at 16 and 31 iterations for lower sum rates. Increasing the transmit power or the number of AP antennas yields higher sum rates for all schemes. However,

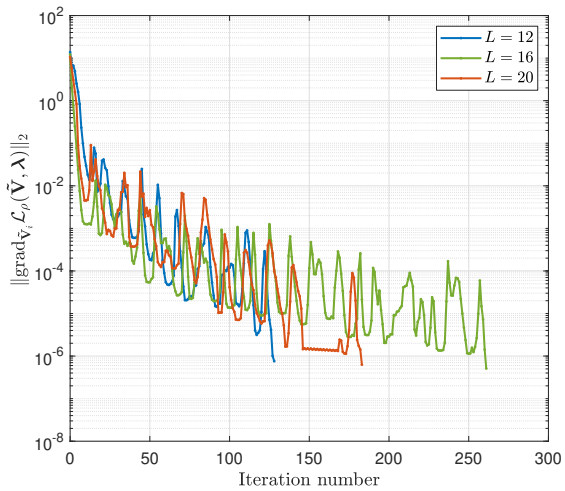


Fig. 3: Convergence rate of the RCG with different numbers of AP antennas.

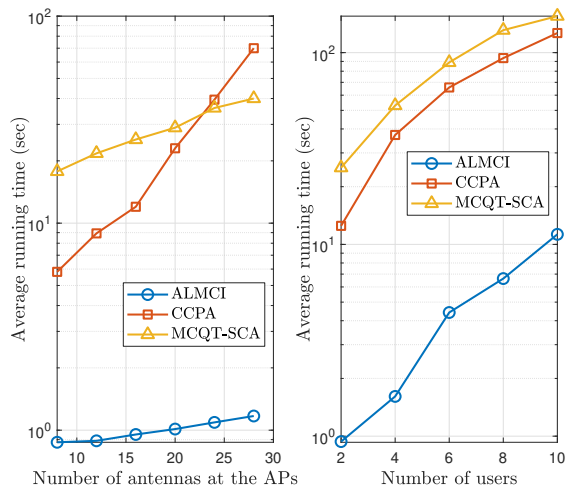


Fig. 4: Average running time versus number of users and AP antennas.

the proposed ALMCI converges in 4 iterations regardless of simulation settings. The other two algorithms need at least 15 and 31 iterations to converge yet achieve lower sum rates. These findings highlight the ability of ALMCI to optimize performance in various environments rapidly.

Fig. 3 illustrates the norm of the gradient of the Lagrangian function, $\|\mathrm{grad}_{\tilde{\mathbf{V}}_i}\mathcal{L}\rho(\tilde{\mathbf{V}}, \boldsymbol{\lambda})\|_2$, as it evolves across iterations for various numbers of antennas. Initially, the gradient norm decreases rapidly for all configurations, indicating that the optimization algorithm quickly approaches the optimal regions characterized by lower gradient norms. As the iterations progress, this decrease becomes more gradual, and fluctuations become more frequent. They indicate the algorithm's step size and direction adjustment based on gradient guidance. Notably, the amplitude of these fluctuations becomes larger with higher values of L. This increase arises from the challenges of managing a higher-dimensional space, highlighting the impact of antenna numbers on the optimization's behavior.

Fig. 4 plots the average running times for the ALMCI, CCPA, and MCQT-SCA algorithms, evaluating their performance across varying numbers of AP antennas and different numbers of users. The first graph reveals that the average running times for CCPA and MCQT-SCA increase with the

TABLE II: Comparison of convergence.

	$L = 8$, $p_{\max} = 25\,\text{dBm}$		$L = 16$, $p_{\max} = 25\,\text{dBm}$		$L = 8$, $p_{\max} = 30\,\text{dBm}$		$L = 16$, $p_{\max} = 30\,\text{dBm}$	
	Sum rate (bps/Hz)	Iterations	Sum rate (bps/Hz)	Iterations	Sum rate (bps/Hz)	Iterations	Sum rate (bps/Hz)	Iterations
ALMCI	26.7211	4	28.9003	4	30.0492	4	32.1721	4
CCPA	19.0686	16	23.6679	19	23.2935	19	26.5908	15
MCQT-SCA	25.0315	31	28.1321	31	27.0484	31	29.9797	31

TABLE III: Comparison of memory usage.

Method	$M = 2$	$M = 3$	$M = 4$
ALMCI	0.29 MB	20.59 MB	29.71 MB
CCPA	0.79 GB	3.81 GB	12.04 GB
MCQT-SCA	10.74 MB	28.56 MB	37.20 MB

antenna number. These trends reflect the algorithms' iterative nature and the computational intensity of the SCA/SDR process, increasing computational complexity as the antenna array expands. In contrast, ALMCI exhibits a more modest and gradual increase, highlighting its computational efficiency and ability to exploit larger antenna arrays. For example, with 4 APs having 16 antennas and 30 dBm transmit power, ALMCI has 12 and 27 times faster algorithm execution time compared to the CCPA and MCQT-SCA algorithms, respectively.

The second graph shows the average run time as a function of the number of users. All algorithms show a rising trend; however, CCPA and MCQT-SCA increase sharply, i.e., poor scalability with user growth. Because CCPA and MCQT-SCA are required to solve larger optimization problems and execute more complex matrix operations. Conversely, ALMCI demonstrates a slower increase in running time, indicating that its MO approach scales more effectively and handles additional users more efficiently.

The ALMCI's reduced execution time and computational complexity stem from its search in a manifold with $(L+1)KM$ dimensions, compared to the larger Euclidean space of $MLKN$ dimensions used by conventional methods like CCPA and MCQT-SCA. This dimensionality reduction simplifies gradient computations, step size tuning, memory usage, and numerical stability, eliminating the need for explicit constraint handling and iterative feasibility checks. ALMCI's use of Riemannian gradients and manifold curvature provides precise updates and large steps, enabling fast convergence [13], [14], [16], [49]. These features make it ideal for real-time ISAC systems in large and dynamic networks.

Table III compares memory usage in MB and GB for three values of M. Memory usage increases with M for all methods. CCPA consistently uses significantly more memory than ALMCI and MCQT-SCA, with ALMCI using the least. For $M = 4$, ALMCI uses only 0.25% of the memory required by CCPA and 78% compared to MCQT-SCA. Thus, the proposed algorithm offered massive memory savings compared to CCPA.

Memory-intensive algorithms, such as CCPA, significantly increase energy consumption due to the power required for larger memory access and maintenance. This impacts both dynamic power (from frequent memory operations) and static power (from leakage currents in larger memory arrays) [50]. Elevated energy consumption results in greater heat dissipation, complicating thermal management and potentially necessitating advanced cooling mechanisms, adding to the power budget and chip complexity. Memory also plays a critical role in chip costs. High memory requirements, such as those of CCPA compared to ALMCI or MCQT-SCA that needs only a few MBs, substantially increase material and manufacturing costs [51].

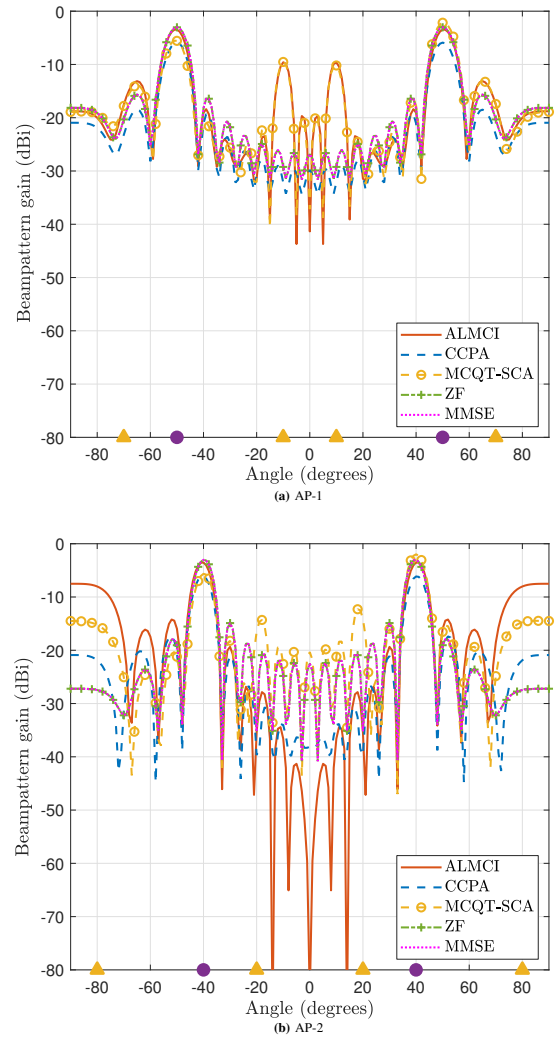
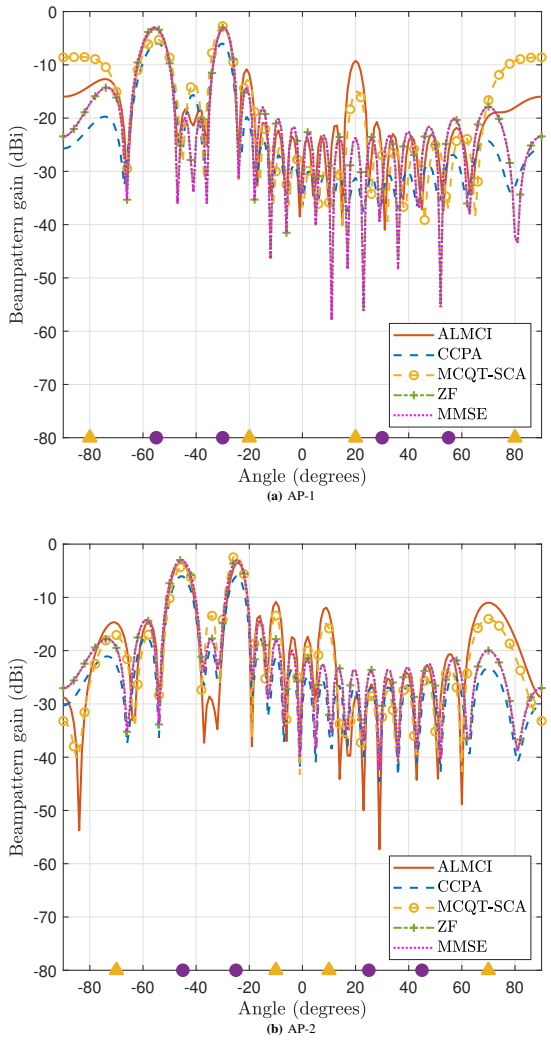


Fig. 5: Directional gain profiles for various algorithms over a $\pm 90^\circ$ angular spread with $K = 2$ user, $N = 4$ targets, and $L = 20$ antennas.

D. Sensing Performance

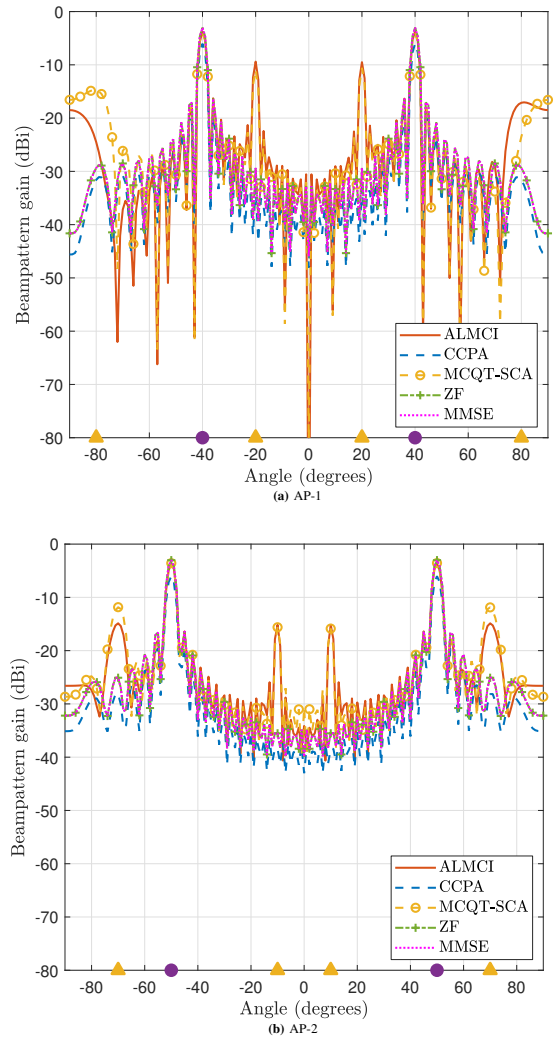
Fig. 5 plots the directional gain profiles with $K = 2$ users and $N = 4$ targets. The subfigures (a) and (b) are for AP-1 and AP-2, respectively. The ALMCI algorithm designs the beam shape with high precision to match the user distribution and the sensing environment. It shows sharp, distinct peaks patterning



(a) AP-1

(b) AP-2

Fig. 6: Directional gain profiles for various algorithms over a $\pm 90^\circ$ angular spread with $K = 4$ user and $N = 4$ targets, and $L = 20$ antennas.



(a) AP-1

(b) AP-2

Fig. 7: Directional gain profiles for various algorithms over a $\pm 90^\circ$ angular spread with $K = 2$ user and $N = 4$ targets and $L = 50$ antennas.

a strong directional focus. This is expected as MO exploits the geometric structure of the problem to achieve finer beam steering and width control, which minimizes interference and maximizes signal strength in the desired directions. The CCPA and MCQT-SCA algorithms, however, demonstrate broader peaks. This beampattern spread may result from relaxing the non-convex problem into a convex one and the concomitant loss of optimality. Such broader beams may not be as effective in null-steering or avoiding interference as ALMCI, negatively impacting user channels when closely spaced.

The conventional ZF/MMSE algorithms offer baselines for comparison. The ZF ideally creates deep nulls in non-intended directions. However, this inversion can lead to power amplification issues when dealing with channels that are near singular or when noise enhancement occurs. The MMSE approach introduces a compromise between inverting the channel and minimizing the power of the transmitted signal. Ultimately, the choice of algorithm depends on the specific requirements of the communication system or sensing application. The accuracy and low computational complexity of the ALMCI in beamforming are suitable for a large number of users and target scenarios with strict interference constraints. In contrast, ZF and MMSE yield simplicity and reduced computational demands in less challenging environments.

Fig. 6 showcases the directional gain with increased users $K = 4$. The directional angles for users from AP-1 and AP-2 are selected as $\{-55, -30, 30, 55\}^\circ$ and $\{-45, -25, 25, 45\}^\circ$, respectively. For AP-1, the ALMCI algorithm's gain profile is characterized by its precision in peak alignment with user locations, indicating an effective concentration of signal power in the desired directions. This precise alignment can lead to enhanced signal reception for the intended users. Again, the CCPA and MCQT-SCA algorithms show broader peaks that result from their dependence on SCA/SDR optimizations. These broader beams can indicate a potential for spillover effects where signal power is distributed over a wider angular range, and is less efficient than ALMCI. The beamforming pattern of ZF/MMSE strikes a balance between interference reduction and gain maximization; their effectiveness is highly dependent on the channel conditions, and their performance degrades when users are closely spaced.

For AP-2, similar trends are observed, although the magnitude and sharpness of the peaks appear slightly varied, reflecting the impact of the AP's positioning relative to users. This change suggests that environmental factors and user dis-

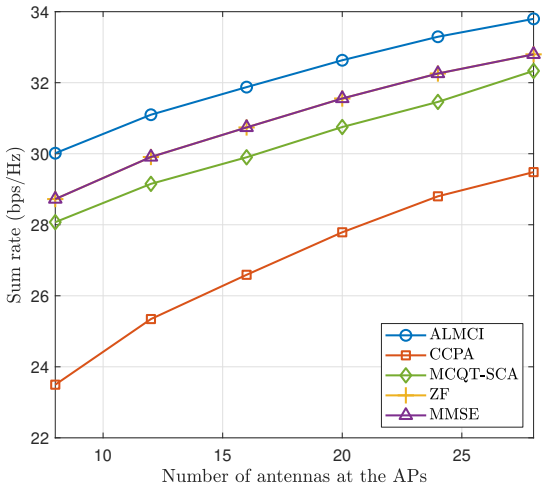


Fig. 8: Sum rate versus the number of AP antennas, L.

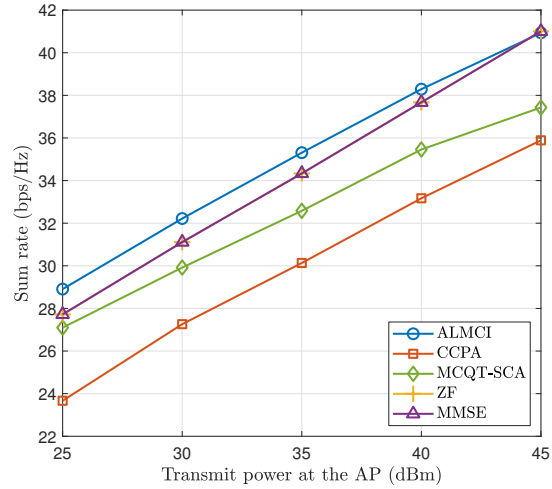


Fig. 9: Sum rate versus the AP transmit power, $p_{\max}$.

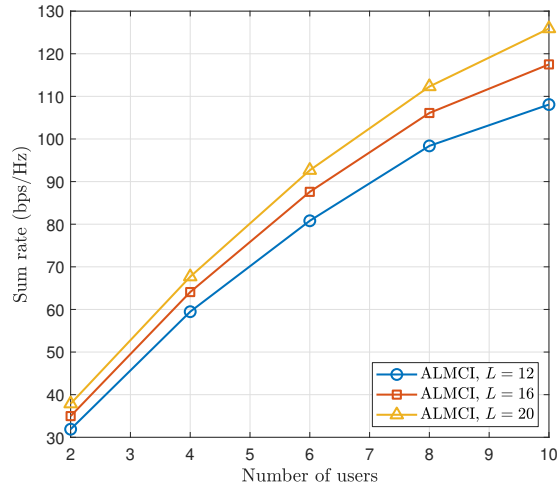


Fig. 10: Sum rate versus the number of users, K.

tribution considerably influence the algorithms' performance. In both AP scenarios, more users introduce optimization challenges, such as a high spatial user distribution and a more complex interference pattern. As ALMCI maintains high precision in its beamforming, it can meet those demands.

In Fig. 7, directional gain profiles of the algorithms are illustrated for the number of antennas $L = 50$. This increase in antennas affords finer control over beam shaping, clearly evidenced in the gain profiles for both AP-1 and AP-2. The ALMCI algorithm distinguishes itself, especially at AP-1, by delivering highly focused beam peaks. These sharp peaks indicate its precision in directing energy toward intended targets, significantly enhancing peak gains for sensing targets compared to other algorithms. Such precise beam direction could translate into improved spatial resolution for sensing applications and elevated data rates for communication users due to reduced inter-user interference.

The ALMCI algorithm reduces sidelobes (i.e., interference) relative to other algorithms. Thus, its efficacy in minimizing potential interference is evident. Conversely, the gain profiles of the other two lack the same peak sharpness as ALMCI, suggesting a less targeted energy distribution. This indicates that CCPA and MCQT-SCA might not fully exploit the spatial multiplexing benefits of the antenna array. The ZF and MMSE algorithms align the beampattern gains toward the communication users without sensing the targets. Overall, expanding the antenna array magnifies the capabilities of all algorithms.

E. Communication Performance

Fig. 8 compares the achieved sum rates as a function of the number of AP antennas with $p_{\max} = 30\,\mathrm{dBm}$. While all the algorithms can leverage the spatial multiplexing gains, ALMCI consistently outperforms the others, showing substantial gains as L increases. Specifically, at $L = 12$, it secures $22.7\,\%$ and $6.7\,\%$ higher sum rates than CCPA and MCQT-SCA, and a $4.0\,\%$ gain over ZF and MMSE. Note that ZF and MMSE have virtually identical performance as the system operates in a high transmit power regime with a low noise variance, i.e., $\sigma^2 = -80\,\mathrm{dBm}$. Conversely, CCPA and MCQT-SCA perform poorer due to approximations and rank relaxations.

Fig. 9 depicts the sum rate as a function of the AP transmit power, $p_{\max}$. Once again, ZF and MMSE perform identically. ALMCI consistently achieves the highest sum rate. For instance, at $p_{\max} = 30\,\mathrm{dBm}$, it surpasses the CCPA, MCQT-SCA and ZF/MMSE by $18.2\,\%$, $7.7\,\%$, and $3.6\,\%$, respectively. All algorithms increase the sum rate with rising transmit power. Unlike ZF and MMSE, ALMCI considers sensing targets, contributing to higher power efficiency and more significant rate gains per power unit. However, CCPA and MCQT-SCA are less efficient in converting increased power to rate gains.

Fig.10 shows the sum rate of ALMCI as a function of the number of users for different antenna numbers per AP. Only ALMCI is included due to its lower time and computational complexity than CCPA and MCQT-SCA, whose execution times increase significantly with the number of users (Fig.4). For instance, generating a simulation with many users can take about a week, as complexity grows exponentially with the number of users, APs, or antennas.

For any number of antennas per AP, the sum rate of the ALMCI algorithm increases with the number of users, demonstrating its efficiency in handling additional users. Furthermore, more antennas per AP increases the sum rate for a

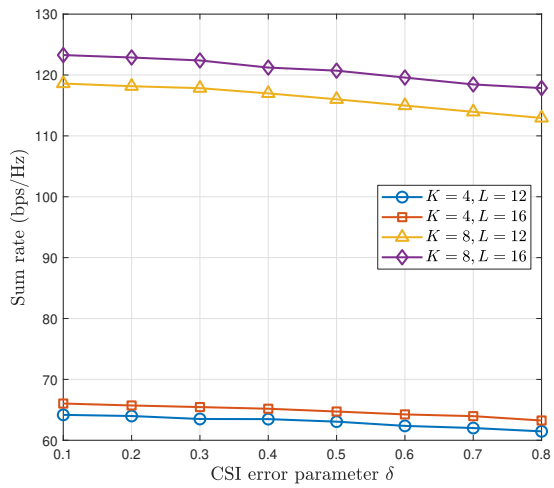


Fig. 11: Sum rate as a function of CSI imperfection δ.

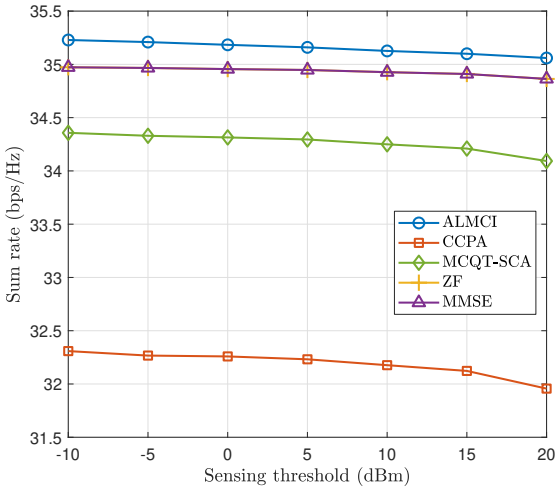


Fig. 12: Sum rate versus the sensing beampattern gain threshold.

given number of users, highlighting the benefits of increased antenna count in managing and distributing data effectively.

F. CSI Impairments

Fig. 11 studies the impact of CSI imperfections/errors on the proposed ALMCI algorithm. This is particularly important as the CSI errors adversely affect the wireless system performance and reliability, including signal reception and beamforming [52], [53]. In Fig. 11, the true channel x is modeled as $\tilde{x} = x + e$, $x \in \{[\mathbf{h}_{mk}]_l\}$ for $l \in \{1, \ldots, L\}$, $\tilde{x}$ denotes the estimated channel, and e is estimation noise modeled as $e \sim \mathcal{N}(0, \sigma_e^2)$ [52]–[54]. Here, the error variance plays a crucial role in quantifying the quality of channel estimation [52]–[54]. We model it as $\sigma_e^2 = \delta|x|^2$, where $|x|$ denotes the magnitude of the true channel and $0 \leq \delta \leq 1$ is a measure of CSI error magnitude.

Fig. 11 shows the relationship between the sum rate and the CSI error parameter, δ. The sum rate remains high with perfect CSI ($\delta \to 0$). The sum rate declines as $\delta \to 1$ due to mismatches between actual and estimated channels.

G. Communication-Sensing Trade-off

Fig. 12 illustrates the trade-off by comparing the sum rates of various algorithms across sensing thresholds. Under per-AP power constraints, higher sensing thresholds prioritize sensing requirements, reducing the sum rate. ALMCI achieves the highest sum rate, CCPA the lowest, while MCQT-SCA performs moderately. ZF and MMSE show consistent but lower sum rates than ALMCI.

VII. Conclusion

The concept of CF-ISAC is still in its early stages, with the literature being relatively nascent. As a result, current CF-ISAC beamforming algorithms are suboptimal and may not scale well due to their failure to leverage the geometric characteristics of the problem. This paper addresses these limitations by developing a novel beamforming algorithm to maximize the communication sum rate while meeting sensing gain targets and AP power constraints. The key ideas are restricting the search to a complex oblique manifold and augmenting the objective function with a penalty term. These ensure the search is on a smooth manifold and transmit power violations (8b) are handled. The proposed ALMCI algorithm enhances beamforming precision and improves the sum rate and spatial resolution. Extensive simulation shows its superior performance compared to existing state-of-the-art methods.

Specifically, it is evaluated against CCPA [2]–[4] and MCQT-SCA [5] methods. It notably slashed computational/time complexity and memory usage while boosting both sensing gain and sum rate. These improvements result in lower power consumption and hardware costs, particularly in large-scale networks, i.e., the reduced computational load on network hardware reduces energy consumption and cooling costs, extending hardware lifespan, improving resource utilization, and minimizing maintenance. They also enable fast response to dynamic environments, allowing the network to support more concurrent users/targets [55]. Conversely, by leveraging problem geometry and Lagrangian penalty for constraint management, the ALMCI algorithm lays a solid foundation for efficiently tackling intricate optimization challenges, rendering it ideal for burgeoning large-scale networks.

As CF-ISAC research is still in its early stages with limited literature [2]–[7], our work advances the field by introducing a novel beamforming approach. However, several open challenges remain. Accurate channel estimation is crucial for both beamforming and interference management in communication, and for precise target parameter estimation in sensing, necessitating adaptive estimation techniques. Limited fronthaul capacity can degrade communication and sensing performance by restricting data transmission and target estimation, reducing throughput and accuracy. Efficient AP selection, user/target clustering, and resource allocation strategies are needed to balance coverage, data rates, and sensing accuracy. Additionally, synchronization errors, such as time and phase misalignment, can degrade communication (e.g., SINR) and sensing (e.g., target detection accuracy), requiring robust synchronization protocols tailored for CF networks.

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Shayan Zargari received the B.Sc. degree in Electronic Engineering from Azad University, South Tehran Branch, Tehran, Iran, in 2018, and the M.Sc. degree in Telecommunication Engineering from the Iran University of Science and Technology, Tehran, Iran, in 2020. He is currently pursuing the Ph.D. degree in communications and signal processing at the University of Alberta, Edmonton, AB, Canada.

From 2019 to 2020, he was a visiting researcher at the Electronics Research Institute, Sharif University of Technology, Tehran, Iran, and from 2020 to 2021, he was a visiting researcher at the Department of Electrical and Computer Engineering, Tarbiat Modares University, Tehran, Iran. His research interests include optimization theory, integrated sensing and communication (ISAC), backscatter communication (BackCom), intelligent reflecting surfaces (IRS), unmanned aerial vehicle (UAV) communications, resource allocation in fifth-generation (5G)/six-generation (6G) wireless communication, and green communication. He serves as a reviewer for several IEEE journals, including IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, IEEE TRANSACTIONS ON COMMUNICATIONS, and IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS.

Diluka Galappaththige (S'17–M'22) is a postdoctoral research fellow at the Department of Electrical and Computer Engineering, University of Alberta, Canada. He received his B.Sc. degree (First-class honor) in Electrical and Electronic Engineering from the Department of Electrical and Electronic Engineering, University of Peradeniya, Sri Lanka, in 2017, and his Ph.D. in Electrical and Computer Engineering from the School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University, Carbondale, IL, USA, in 2021.

Dr. Galappaththige has been awarded a post-doctoral fellowship from NSERC for the academic year 2024-2026. His current research interests include, but are not limited to, the design, modeling, and analysis of massive multiple-input multiple-output (mMIMO) communication (i.e., including co-located mMIMO and cell-free/distributed mMIMO), full-duplex communication (FD), backscatter communication (BackCom), reconfigurable intelligent surfaces (RISs), integrated sensing and communication (ISAC), near-field ISAC, wireless power transfer, and emerging technologies for enabling fifth-generation (5G) and beyond wireless networks. He was a recipient of the Exemplary Reviewer Award for IEEE Wireless Communications Letters (TWCL) in 2020, and IEEE Communications Letters (TCL) in 2021. He has actively served as a reviewer for a variety of IEEE journals and conferences, including IEEE Transactions on Communications, IEEE Internet of Things Journal, IEEE Transactions on Vehicular Technology, IEEE Transactions on Green Communications and Networking, IEEE Access, IEEE Network, IEEE Open Journal of Communications Society, TCL, TWCL, IEEE International Conference on Communications, and IEEE Global Communications Conference.

Chintha Tellambura (Fellow, IEEE) received the B.Sc. degree (First class) in electrical and electronic engineering from the University of Moratuwa, Sri Lanka, the M.Sc. degree in electrical engineering from King's College, University of London, and the Ph.D. degree in electrical engineering from the University of Victoria, Canada. He was with Monash University, Australia, from 1997 to 2002. Dr. Tellambura is a Professor in the Department of Electrical and Computer Engineering at the University of Alberta. He has authored or co-authored over 650 journals and conference papers, demonstrating his expertise in the field. According to Google Scholar, his exceptional scholarly contributions have earned him an impressive H-index of 84. Dr. Tellambura has made significant contributions to various areas of research, including future wireless networks, machine learning for wireless networks, and signal processing.

Recognizing his outstanding accomplishments, he was elected as an IEEE Fellow in 2011 for his noteworthy contributions to physical layer wireless communication theory. In 2017, he was further honored as a fellow of the Canadian Academy of Engineering, a testament to his exceptional achievements. His dedication and expertise have been acknowledged through prestigious awards, including the Best Paper Awards in the Communication Theory Symposium in 2012, the IEEE International Conference on Communications (ICC) held in Canada in 2017, and another ICC in France. Moreover, Dr. Tellambura has been honored with the esteemed McCalla Professorship and the Killam Annual Professorship by the University of Alberta, further underscoring his significant impact on academia. Dr. Tellambura has also played a vital role in editorial responsibilities within the IEEE community. He served as an Editor for the IEEE Transactions on Communications from 1999 to 2011 and for the IEEE Transactions on Wireless Communications from 2001 to 2007. In the latter role, he was Area Editor of Wireless Communications Systems and Theory from 2007 to 2012, contributing to advancing the field through his editorial expertise.

Geoffrey Ye Li (Fellow, IEEE) is currently a Chair Professor at Imperial College London, UK. Before joining Imperial in 2020, he was a Professor at Georgia Institute of Technology, USA, for 20 years and a Principal Technical Staff Member with AT&T Labs – Research (previous Bell Labs) in New Jersey, USA, for five years. He made fundamental contributions to orthogonal frequency division multiplexing (OFDM) for wireless communications, established a framework on resource cooperation in wireless networks, and introduced deep learning to communications. In these areas, he has published around 700 journal and conference papers in addition to over 40 granted patents. His publications have been cited over 75,000 times with an H-index of 127. He has been listed as a Highly Cited Researcher by Clarivate/Web of Science almost every year.

Dr. Geoffrey Ye Li was elected to Fellow of the Royal Academic of Engineering (FREng), IEEE Fellow, and IET Fellow for his contributions to signal processing for wireless communications. He received 2024 IEEE Eric E. Sumner Award, 2019 IEEE ComSoc Edwin Howard Armstrong Achievement Award, and several other awards from IEEE Signal Processing, Vehicular Technology, and Communications Societies.